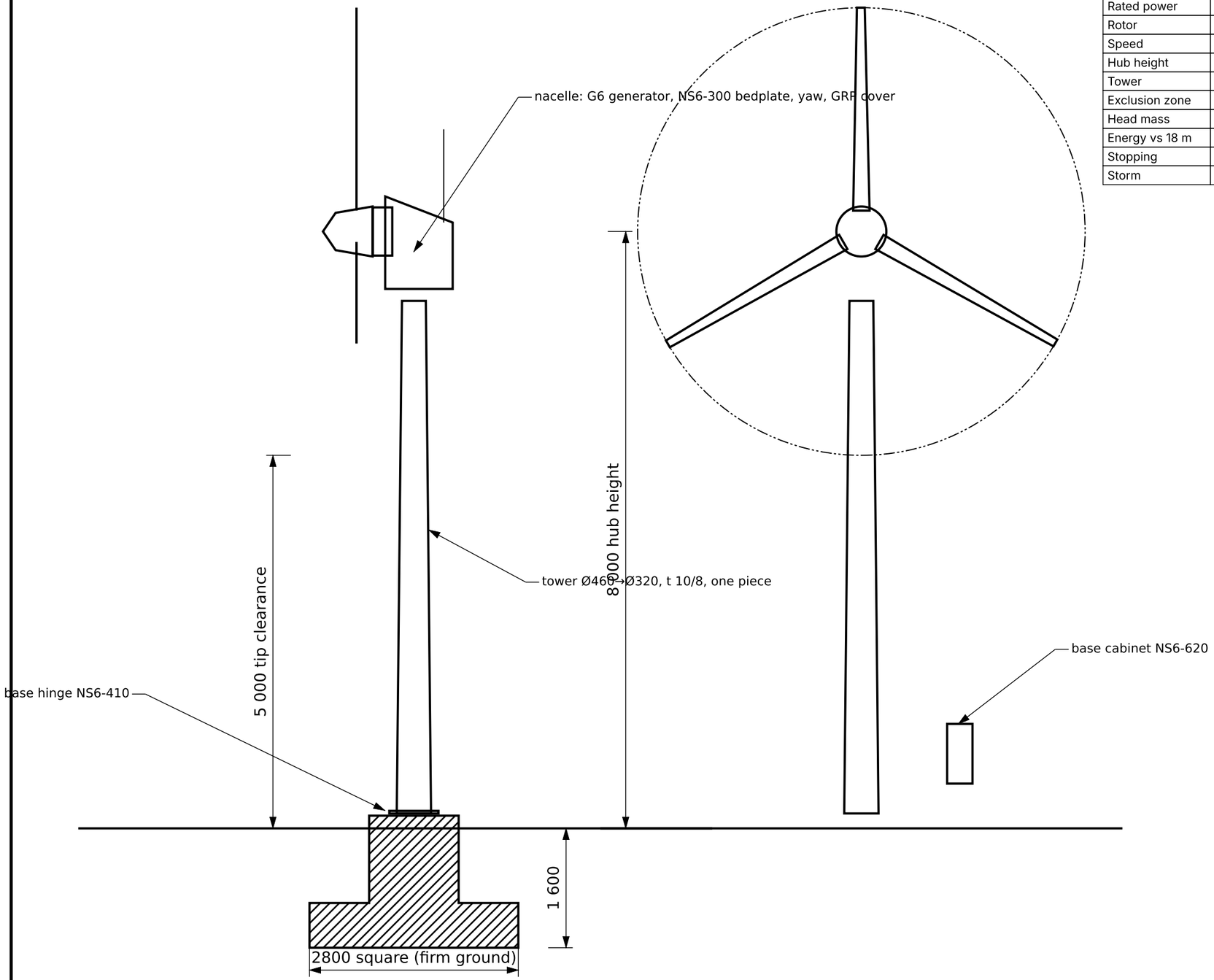
SIDE VIEW (wind from the left)



VIEW FROM UPWIND

Item	Value
Rated power	6 kW electric at ≈ 9.7 m/s (hub)
Rotor	Ø6.0 m, 3 blades, TSR 5.5, pitch −2° fixed
Speed	41–164 rpm, trip 189 rpm
Hub height	8.0 m; lowest tip 5.0 m
Tower	661 kg, f1 3.5 Hz (3.28 soft+ice)
Exclusion zone	60–77 rpm (3P × f1)
Head mass	982 kg above the tower flange (CAD)
Energy vs 18 m	−18 % coast … −26 % inland
Stopping	K1 resistor 3 × 4 Ω + spring brake 1.5 kNm
Storm	stop, yaw 90° out, hold

AXONOMETRIC VIEW (not to scale)



NOTES

1. Docs 28–34. Rev B: hub height 8 m (was 18 m), tower one piece, footing re-sized; all new parts drawn (index NS6-001).
2. Keep people and vehicles out of the rotor disc: fence or marked zone of radius 4 m round the tower base; ice-fall zone 1.5 × (H + D) = 21 m when icing is possible.
3. Siting at 8 m: rotor bottom must be clear of the wake of buildings and shelter belts (≥ 20 × obstacle height upwind in the main wind directions).

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

GENERAL ARRANGEMENT · TURBINE ON 8 m TOWER

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-000	B · 2026-11-06	1:60
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see BOM NS6-002/003	≈ 982 kg head + 661 kg tower	Claude / not checked

Drawing	Title	Source
NS6-000	GENERAL ARRANGEMENT · TURBINE ON 8 m TOWER	S6 rev B
NS6-001	DRAWING INDEX	S6 rev B
NS6-002	BILL OF MATERIALS (1/2)	S6 rev B
NS6-003	BILL OF MATERIALS (2/2)	S6 rev B
NS6-010	GENERAL ARRANGEMENT · NACELLE SECTION	S6 rev B
NS6-020	ROTOR ASSEMBLY · HUB, BLADES, SPINNER	S6 rev B
NS6-100	BLADE · PLANFORM, TWIST AND ROOT	S6 rev B
NS6-101	BLADE · SECTIONS AND LAMINATE SCHEDULE	S6 rev B
NS6-105	BLADE ROOT INSERT M12 (BONDED)	S6 rev B
NS6-110	BLADE MOULD SET (PLUG, TWO HALVES, FRAME)	S6 rev B
NS6-115	ROOT INSERT FIXTURE AND WEB MOULD (TOOLS)	S6 rev B
NS6-130	PITCH SETTING · ROOT SLOTS, WASHER PLATES, GAUGE	S6 rev B
NS6-200	FIXED HUB WELDMENT	S6 rev B
NS6-210	SPINNER (GRP) AND BRACKETS NS6-211	S6 rev B
NS6-300	BEDPLATE WELDMENT 6 kW (LIGHT)	S6 rev B
NS6-310	CALIPER BRACKET NP1-310 AND YAW DRIVE BRACKET NP1-420	S6 rev B
NS6-400	TOWER 8 m · ONE-PIECE TILT-UP MONOPOLE	S6 rev B
NS6-410	BASE HINGE BRACKETS AND PIN	S6 rev B
NS6-420	GIN POLE AND TILT-UP RIGGING (ERECTION TOOL)	S6 rev B
NS6-500	FOUNDATION · SPREAD FOOTING AND ROCK PLINTH (8 m)	S6 rev B
NS6-510	ANCHOR CAGE AND TEMPLATE	S6 rev B
NS6-600	BRAKING RESISTOR BOX AND SAFETY CHAIN	S6 rev B
NS6-610	G6 PLAIN REAR COVER (NO FAN BLADES)	S6 rev B
NS6-620	TOWER BASE CABINET, STAND AND RESISTOR BOX POSITION	S6 rev B
NS6-700	NACELLE COVER · LOWER TRAY AND UPPER LID (GRP)	S6 rev B
NS6-710	COVER ARMS, REAR RAIL, MET MAST NS6-720, JUNCTION BOX NS6-730	S6 rev B
NS6-800	CABLING ROUTES, EARTHING AND LIGHTNING	S6 rev B
NS6-900	FASTENER AND TORQUE SCHEDULE	S6 rev B
NS6-910	BOUGHT PARTS SPECIFICATION (ENVELOPES AND INTERFACES)	S6 rev B
NP1-100	SPINDLE (STATIONARY MAIN SHAFT)	12 kW NP1 rev B
NP1-101	FRONT SPACER + REAR SEAL SLEEVE NP1-102	12 kW NP1 rev B
NP1-211	MAIN BEARING SEAL COVERS (FRONT / REAR)	12 kW NP1 rev B
NP1-230	BRAKE DISC AND ROTOR LOCK HOLES	12 kW NP1 rev B
NP1-320	ROTOR LOCK PIN AND HOLDER	12 kW NP1 rev B
NP1-T01	DRUM ASSEMBLY GUIDE (MAGNET SAFETY TOOL)	12 kW NP1 rev B
NP1-G00	GENERATOR REV C · ASSEMBLY SECTION (G6: no fins, ribs, fan blades)	Rev C (doc 23)
NP1-200C	STATOR CARRIER WITH 45 DOVETAIL GROOVES	Rev C
NP1-201C	STATOR TOOTH SEGMENT LAMINATION	Rev C
NP1-202	STATOR FRONT CLAMP RING	Rev C
NP1-203C	COIL ON SEGMENT (WOUND SEGMENT)	Rev C
NP1-210C	ROTOR DRUM WELDMENT (G6: omit fins and ribs)	Rev C
NP1-220E	MAGNET BLOCK 30 × 15 × 12 (PURCHASE)	Rev C
NP1-221C	STEPPED POLE COMB BAR	Rev C
NP1-222C	MAGNET END RINGS	Rev C
NP1-223C	ROTOR MAGNET PACK	Rev C
NY-000	YAW SYSTEM · GENERAL ARRANGEMENT	doc 25
NY-400	TOWER-TOP ADAPTER WITH FRICTION TRACK	doc 25
NY-410	FRICTION PAD UNIT	doc 25
NY-430	YAW LOCK (SERVICE)	doc 25
NY-440	YAW ENCODER AND CAM LIMIT SWITCH UNIT	doc 25
NY-460	CABLE HANGER AND TWIST LOOP	doc 25

NOTES

1. Rev B (2026-11-06): 8 m tower; every S6 part drawn with an axonometric view; BOM NS6-002/003; fasteners NS6-900; cabling NS6-800.
2. Reused 12 kW drawings are used unchanged except where the BOM says 'G6 variant'. Their PDFs follow in the build book in the order of this index.
3. All sheets NOT CHECKED: prototype release only after review, FE checks (hub arm welds, bedplate ring seat) and the stall-torque test.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

DRAWING INDEX (S6 SET REV B + REUSED 12 kW DRAWINGS)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-001	B · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
–	–	Claude / not checked

Pos	Part no.	Description	Qty	Make/buy	Material / spec	kg each
		A ROTOR				
1	NS6-100	Blade, glass/epoxy, glass UD caps (B1–B3)	3	make	E-glass/epoxy, PVC core	20.6
2	NS6-105	Root insert M12 (10 per blade, bonded)	30	make	C45E+QT zinc-flake coated	0.3
3	NS6-130	Pitch washer plates 28 × 4 hardened (30)	1	make	C45 hardened 45 HRC, zinc-flake	0.5
4	NS6-200	Fixed hub weldment	1	make	S355NL, welded, stress relieved, machined	79.2
5	NS6-210	Spinner (GRP, 4 mm)	1	make	glass/polyester, gelcoat RAL 9010	8.7
6	NS6-211	Spinner brackets (3)	1	make	S235 galv.	2.0
		B DRIVETRAIN AND GENERATOR G6				
7	NP1-100	Spindle	1	make	42CrMo4+QT forged	56.5
8	NP1-101	Front spacer / seal counterface Ø180 (Rev B)	1	make	42CrMo4+QT, induction hardened	1.4
9	NP1-102	Rear seal sleeve Ø195 on the Ø155 shoulder (Rev B)	1	make	42CrMo4+QT, induction hardened	1.3
10	NP1-200C	Stator carrier with 45 dovetail grooves	1	make		39.9
11	NP1-201C	Stator: 45 tooth-segment lamination stacks	1	make		50.5
12	NP1-202	Stator front clamp ring	1	make	1.4404 (non-magnetic)	2.2
13	NP1-203C	Coils on segments, 45 × 28 turns (end windings shown)	1	make	Cu enamelled Ø1.55 class 200, NKN 0.30	20.0
14	NP1-210C	Rotor drum weldment (G6: no fins, no ribs)	1	make	S355J2+N	91.2
15	NP1-211-F	Front bearing seal cover	1	make	S355J2	1.5
16	NP1-211-R	Rear bearing seal cover, bore Ø198 (Rev B)	1	make	S355J2	1.4
17	NP1-220E	Magnets 200 × N45SH 30×15×12 + comb + sleeve	1	make	NdFeB N45SH	10.7
18	NP1-230	Brake disc Ø880/Ø652 with 12 lock holes (Rev B)	1	make	C45E+N 16 mm	33.6
19	NP1-310	Rotor brake caliper bracket (left, Rev B)	1	make	S355J2+N 25 plate	18.1
20	NP1-320	Rotor lock pin Ø20 (engaged, top)	1	make	42CrMo4+QT	0.5
21	NP1-321	Lock pin holder with 2 overspeed sensor seats	1	make	S355J2	24.5
22	NS6-610	G6 plain rear cover (no fan blades)	1	make	EN AW-5083-H111	7.2
23	NS6-913	Tapered roller bearing 32028 X (front + rear)	2	buy	bought 32028 X	4.0
24	NS6-914	Overspeed sensors M18 inductive (2, bought)	1	buy	bought M18 PNP	0.4
25	NS6-915	Spring-applied brake caliper, 1.5 kNm pack (bought)	1	buy	bought, spring-applied	9.0
		C NACELLE				
26	NS6-300	Bedplate weldment 6 kW (light)	1	make	S355J2+N plates, stress relieved, machined	122.1
27	NS6-700-1	Nacelle cover, lower tray (GRP 5 mm)	1	make	glass/polyester, gelcoat RAL 9010	15.7
28	NS6-700-2	Nacelle cover, upper lid (GRP 5 mm)	1	make	glass/polyester, gelcoat RAL 9010	17.5
29	NS6-710	Cover support arms + rear rail	1	make	S235 galv.	8.3
30	NS6-720	Met mast: anemometer, vane, air terminal	1	make	1.4404 tube 42.4 × 2 + bought sensors	4.5
		D YAW SYSTEM				
31	NP1-420	Yaw drive bracket	2	make	S355J2+N	7.7
32	NS6-911	Yaw slewing ring Dpw 360, z150 m4	1	buy	bought: 42CrMo4, induction hardened teeth	55.0
33	NS6-912	Yaw gearmotor 0.18 kW 1:1680, motor brake 2 Nm	2	buy	bought	19.0
34	NY-400	Tower-top adapter with friction track	1	make	S355J2+N weldment	106.7
35	NY-410	Friction pad unit	2	make	S355 / DIN 2093 springs / organic pads	14.0
36	NY-430	Yaw lock (screw block)	1	make	42CrMo4 / S355	6.5
37	NY-440	Yaw encoder + cam limit switch unit	1	make	S355 + absolute encoder + cam switch	6.0
38	NY-460	Cable hanger tube + loop cleat	1	make	S355 tube 168.3 × 4.5	10.0
		E ELECTRICAL				
39	NS6-600	Braking resistor box 3 × 4 Ω (vented)	1	make	1.4301 box + steel-grid resistors	18.0
40	NS6-620	Tower base cabinet IP55 800 × 600 × 300 on stand	1	make	stand S235 galv.; bought enclosure	95.0
41	NS6-730	Nacelle junction box IP66 (400 × 340 × 150)	1	make	bought: 1.4301 enclosure	9.0
42	NS6-800	Drop cable + base duct (cabling route)	1	make	H07RN-F 4G10 + 2 × LiYCY signal; PE duct Ø11	12.0
		F TOWER				
43	NS6-400	Tower 8 m, Ø460 → Ø320, t 10/8, one piece	1	make	S355J2+N, hot-dip galvanised	697.5
44	NS6-401	Hand-hole cover 200 × 350 × 6 with gasket	1	make	S355 galv.	3.3
45	NS6-410	Base hinge brackets + hinge pin Ø50	1	make	S355J2+N galv.; pin 42CrMo4+QT	37.4
		G FOUNDATION				

NOTES

1. Masses from the CAD model (cad/turbine_6kw/model6.py) for made parts; supplier data or estimates for bought parts. Full BOM with sources: bom6.csv.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

BILL OF MATERIALS (1/2)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-002	B · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see lines	see lines	Claude / not checked

Pos	Part no.	Description	Qty	Make/buy	Material / spec	kg each
46	NS6-500	Spread footing (firm ground) with pedestal	1	make	C30/37 XF4 air-entrained, B500C	13200.0
47	NS6-510	Anchor cage 12 × M24 + 4 hinge anchors	1	make	M24 8.8 HDG threaded rod, S355 template ring	143.5
		H TOOLS AND MOULDS				
48	NS6-110-1	Blade mould, lower (pressure side) half on frame	1	make	GRP tooling laminate 12 mm, vinylester tooli	75.4
49	NS6-110-2	Blade mould, upper (suction side) half	1	make	GRP tooling laminate 12 mm	75.4
50	NS6-110-3	Mould frame (RHS 80 × 80 × 4)	1	make	S235 RHS 80×80×4	155.4
51	NS6-115	Root insert fixture (tool)	1	make	S355, machined	9.2
52	NS6-135	Pitch setting gauge (tool)	1	make	Al 6082 + digital inclinometer	1.6
53	NS6-420	Gin pole 3.0 m (erection tool)	1	make	S355 tube 168.3 × 6.3	94.0
		J FASTENERS, CABLES, SITE MATERIALS				
54	NS6-900-01	Stud M12 × 75 10.9 zinc-flake + nut + washer (blade roots)	30	buy	10.9 zinc-flake	0.12
55	NS6-900-02	Bolt M16 × 70 10.9 (hub to drum)	12	buy	10.9 zinc-flake	0.17
56	NS6-900-03	Bolt M20 × 90 10.9 (spindle to bedplate)	12	buy	10.9 zinc-flake	0.32
57	NS6-900-04	HV set M16 × 110 10.9 (yaw inner ring)	16	buy	EN 14399-4	0.26
58	NS6-900-05	Bolt M16 × 70 10.9 (yaw outer ring, pads, caliper)	28	buy	10.9 zinc-flake	0.17
59	NS6-900-06	HV set M20 × 90 10.9 (adapter to tower)	16	buy	EN 14399-4	0.40
60	NS6-900-07	Nut M24 8.8 HDG + plate washer (base, hinge)	44	buy	8.8 HDG	0.15
61	NS6-900-08	Small fasteners kit M8–M12 A4 (cover, spinner, mast, cabinet)	1	buy	A4-70	2.50
62	NS6-800-W1	Cable H07RN-F 4G10 (generator → box)	3	buy	m	0.60
63	NS6-800-W2	Cable H07RN-F 4G10 (box → loop → hand hole)	12	buy	m	0.60
64	NS6-800-W3	Cable NYY-J 4 × 10 (hand hole → cabinet)	4	buy	m	0.50
65	NS6-800-W4	Cable ÖLFLEX 4G1.5 shielded (brake)	16	buy	m	0.10
66	NS6-800-W5	Cable LiYCY 12 × 0.5 (sensors)	16	buy	m	0.12
67	NS6-800-W6	Cable ÖLFLEX 7G1.5 (yaw motors)	16	buy	m	0.15
68	NS6-800-W7	Earthing conductor Cu 50 mm² + ring earth + 2 rods 3 m	1	buy	Cu	18.00
69	NS6-500-M1	Concrete C30/37 XF4 air-entrained (firm ground)	5.5 m³	buy	ready-mix	
70	NS6-500-M2	Rebar B500C (mesh Ø16/150 top + bottom, pedestal cage)	1	buy	≈ 420 kg	420.00
71	NS6-500-M3	Non-shrink grout ≥ 40 MPa, 30 mm under the base flange	60 l	buy		
72	NS6-600-K1	Contactor K1 3-pole NC (mirror contacts) + safety relay PL d	1	buy		2.00
73	NS6-620-PE	Converter, dump chopper, AC module, 48 V module, controller (d	1	make	in-house electronics	45.00

NOTES

1. Masses from the CAD model (cad/turbine_6kw/model6.py) for made parts; supplier data or estimates for bought parts. Full BOM with sources: bom6.csv.

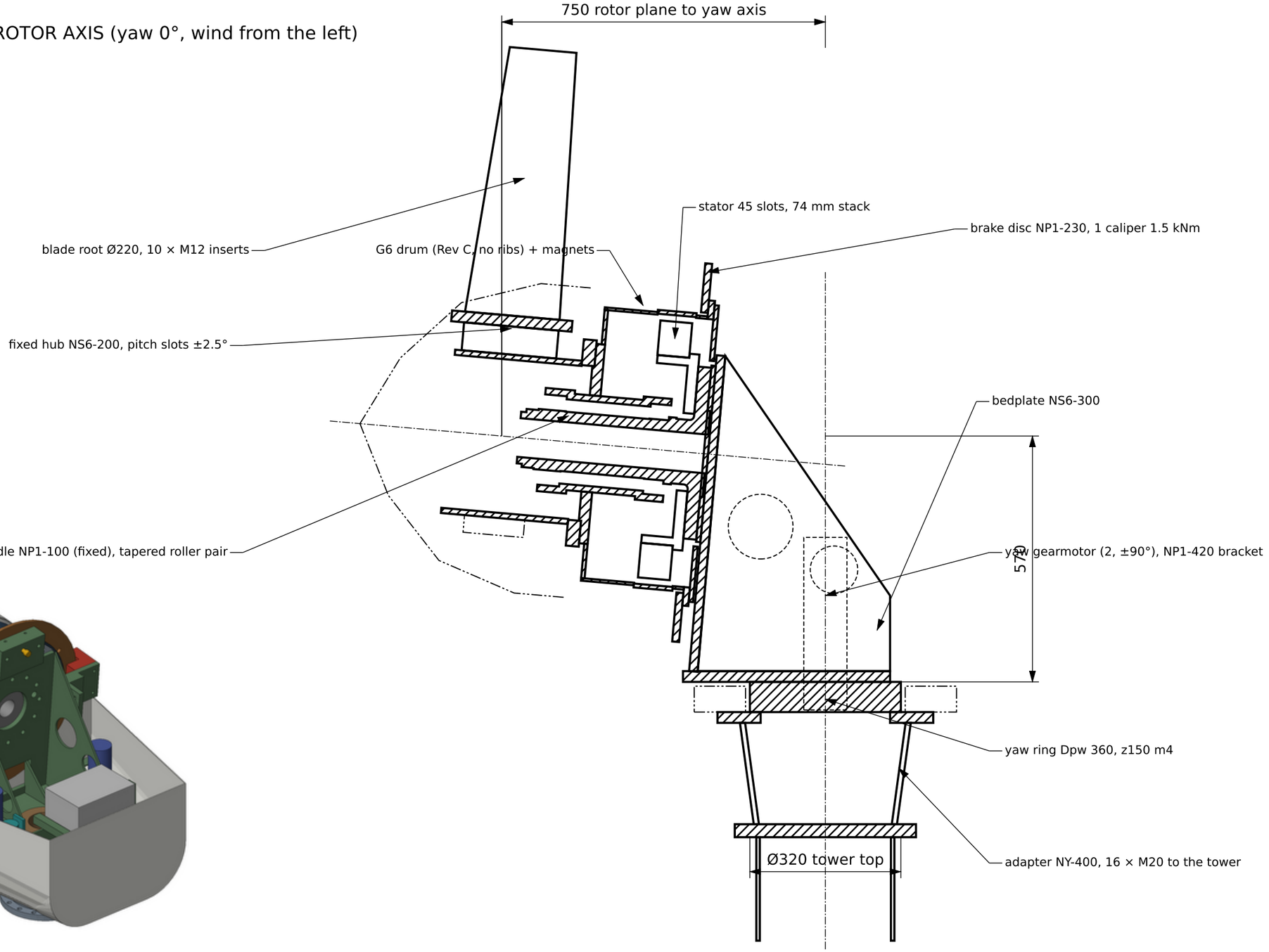
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

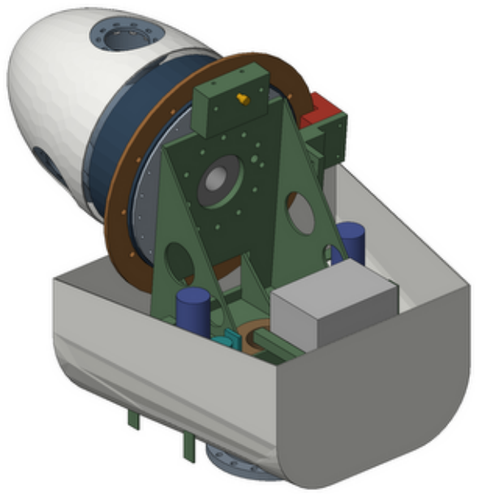
BILL OF MATERIALS (2/2)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-003	B · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see lines	see lines	Claude / not checked

SECTION ON THE ROTOR AXIS (yaw 0°, wind from the left)



AXONOMETRIC VIEW (not to scale)



Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

NOTES

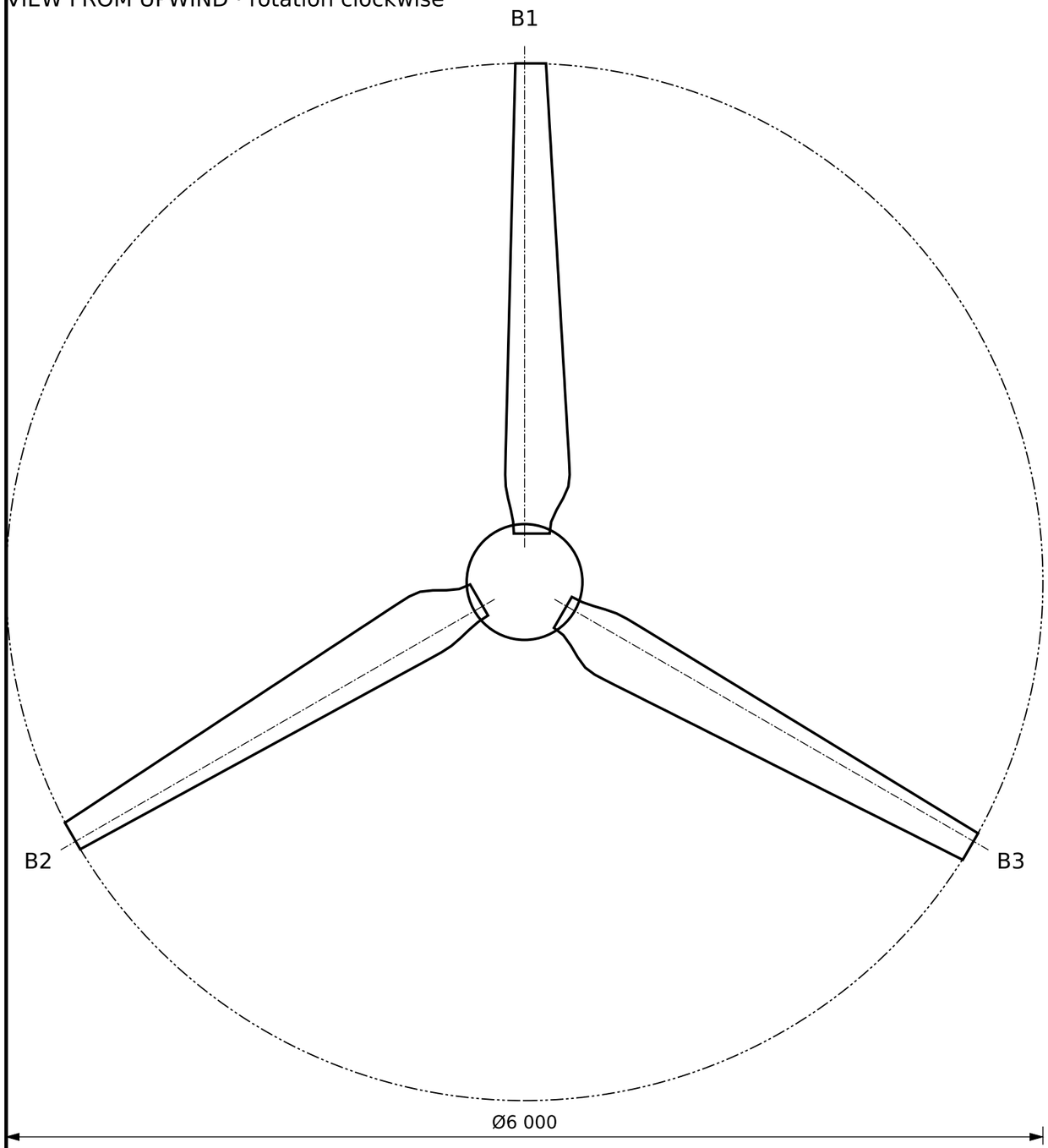
1. Drivetrain geometry from cad/nacelle_p1/geometry.py (Rev B nacelle, Rev C generator, doc 23); fins, ribs and cover blades not fitted on G6 (doc 29).
2. Rotor turns with the drum on two tapered roller bearings on the stationary spindle; the hub bolts to the drum front face (12 × M16 PCD 400).
3. Section shows the hub arm of the upper blade; the other two arms are at ±120°. Spinner (phantom) is GRP, NS6-210. GRP nacelle cover NS6-700 not shown (see NS6-700).
4. Braking resistor box NS6-600 is mounted on the tower base cabinet wall, not in the nacelle (heat). Phase leads run in the hanger tube.

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

GENERAL ARRANGEMENT · NACELLE SECTION

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-010	B · 2026-11-06	1:10
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
see NS6-001	≈ 982 kg above the tower flange	Claude / not checked

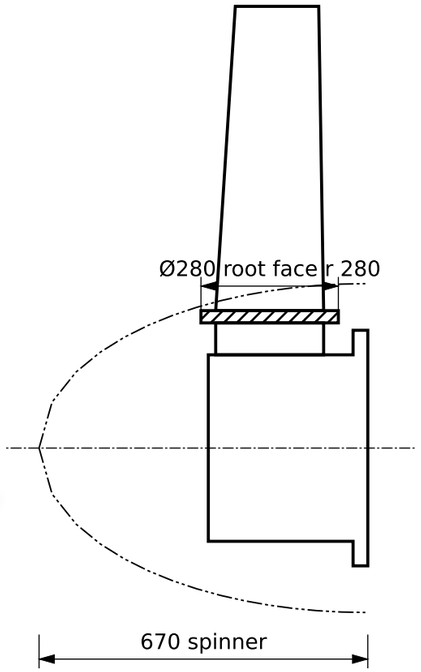
VIEW FROM UPWIND · rotation clockwise



AXONOMETRIC VIEW (not to scale)



SIDE VIEW, upper blade



Item	Value
Blade setting	−2.0° ± 0.1° at r = 2 000 (gauge NS6-135)
Blade set	static moments equal within 0.5 %
Track	tip track within ± 15 mm (string / laser)
Rotor balance	residual ≤ 0.2 kg·m (doc 09 R6.6)
Root bolts	30 × M12 10.9, 95 Nm + 90° (NS6-900)
Hub bolts	12 × M16 10.9 to the drum, 250 Nm

NOTES

1. Blades are numbered B1–B3 at the factory with their static moment; mount B1 on the arm marked 1 (lock-pin side).
2. Set pitch with the blade on the hub, bolts finger tight, gauge on the r = 2 000 station; tighten in a star pattern; re-check after 50 h of operation.
3. Balance the rotor on the turbine with the doc 09 procedure (one trial mass on the spinner brackets).

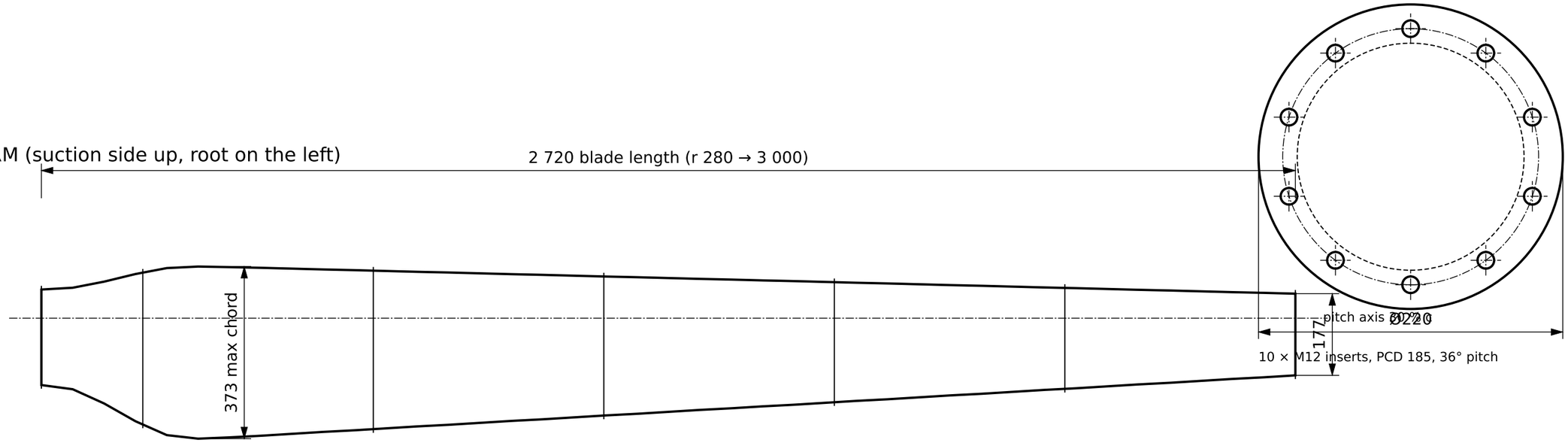
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

ROTOR ASSEMBLY · HUB, BLADES, SPINNER

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-020	B · 2026-11-06	1:30
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see BOM	≈ 162 kg (rotor + hub + spinner)	Claude / not checked

PLANFORM (suction side up, root on the left)



ROOT END (from the hub)

r mm	chord	twist °	t/c	cap mm
280	207	20.0	1.00	4.25
400	254	20.0	0.71	4.25
600	370	20.0	0.32	4.25
800	360	17.1	0.28	4.25
1000	343	13.4	0.26	4.10
1400	310	8.5	0.24	3.38
1800	277	5.5	0.21	2.41
2200	244	3.6	0.19	1.44
2600	210	2.2	0.18	1.00
3000	177	1.2	0.18	1.00

AXONOMETRIC VIEW (not to scale)



NOTES

1. Aerofoil family and twist from analysis/rotor/bem.py (TSR 5.5, cl 1.0, R 3.0 m, root r 0.28 m); twist given at the pitch axis, positive nose up towards feather; blade set at −2° in the arc slots of the hub root flange (NS6-130).
2. Structure doc 30 §2 / doc 10 model: two glass UD caps (table), ±45° glass skins, PVC foam core, single shear web 1.5 mm glass; root wall builds up 12 mm over the first 0.7 m. Cap buckling at minimum thickness governs (use 1.00); strength 0.43 (SLM H), fatigue 0.10 with simulated loads (doc 31).
3. Root: 10 bonded steel inserts M12 NS6-105, PCD 185, pull-out ≥ 60 kN each (test). Bolt force 43 kN at SLM H. Sections and laminate: NS6-101; mould NS6-110.
4. Mass 20.6 kg ± 0.3 kg; static moment matched within 0.5 % per set of three (balance class doc 09). Erosion tape on the outer 1 m of the leading edge.
5. First flap 15.2 Hz, first edge 38.3 Hz at standstill. Tip deflection 0.06 m, tower clearance 0.83 m.

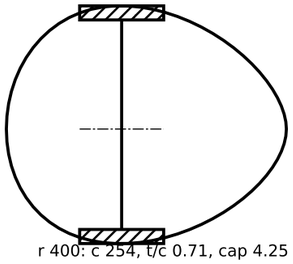
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

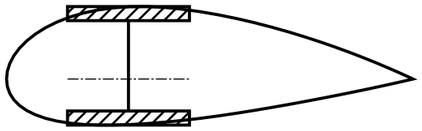
BLADE · PLANFORM, TWIST AND ROOT

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-100	B · 2026-11-06	1:10
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
E-glass/epoxy infused; glass UD caps; PVC core shear web	20.6 kg	Claude / not checked

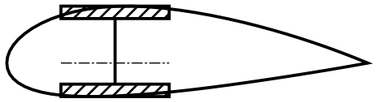
SECTIONS (caps hatched ×3 thickness for clarity, web at 30 % c; LE to the left)



r 400: c 254, t/c 0.71, cap 4.25



r 700: c 368, t/c 0.29, cap 4.25



r 1200: c 327, t/c 0.25, cap 3.75



r 2000: c 260, t/c 0.20, cap 1.93



r 2800: c 194, t/c 0.18, cap 1.00

Zone (r mm)	Shell skin	Core	Spar cap (each side)	Web
280–700 root	2 × 600 g ±45° + 0–12 mm UD/biax build-up	–	merges into root	–
700–1 400	2 × 450 g ±45°	PVC H80 8 mm	UD 1 200 g: 4.25→3.75 mm	2 × 450 g ±45° on 10 mm core
1 400–2 200	2 × 450 g ±45°	PVC H80 6 mm	UD: 3.50→2.25 mm	same
2 200–3 000 tip	2 × 300 g ±45°	PVC H80 5 mm	UD: 2.00→1.00 mm (min)	ends at r 2 900
LE r 2 000–3 000	erosion tape 0.3 mm PU			

NOTES

1. Vacuum infusion, epoxy (Tg ≥ 80 °C, cold-curing then post-cure 8 h at 70 °C), fibre volume ≥ 50 %. Each blade: two shells + web, bonded with structural paste (1–3 mm gaps), leading-edge and trailing-edge flanges 30 mm.
2. Cap thickness follows design6.json t_cap_mm (doc 30 §2); minimum cap 1.0 mm governed by buckling: do not reduce. UD glass E-glass 1 200 g/m² stitched, 0° ± 2° to the pitch axis.
3. Root: laminate builds up from r 700 to a 28 mm wall at the root face around the 10 bonded inserts NS6-105 (fixture NS6-115). Machine the root face square to the pitch axis within 0.2 mm after cure.
4. Gelcoat: polyurethane, white RAL 9010, 0.4 mm. Drain hole Ø4 at the tip trailing edge. Lightning: none on 3 m blades (LPL I, doc 02: tower and nacelle carry it; blade receptor option).
5. Quality: weigh and record mass + static moment; tap test the bonds; one blade per 20 static-tested to SLM H × 1.35 (doc 30).

AXONOMETRIC VIEW (not to scale)

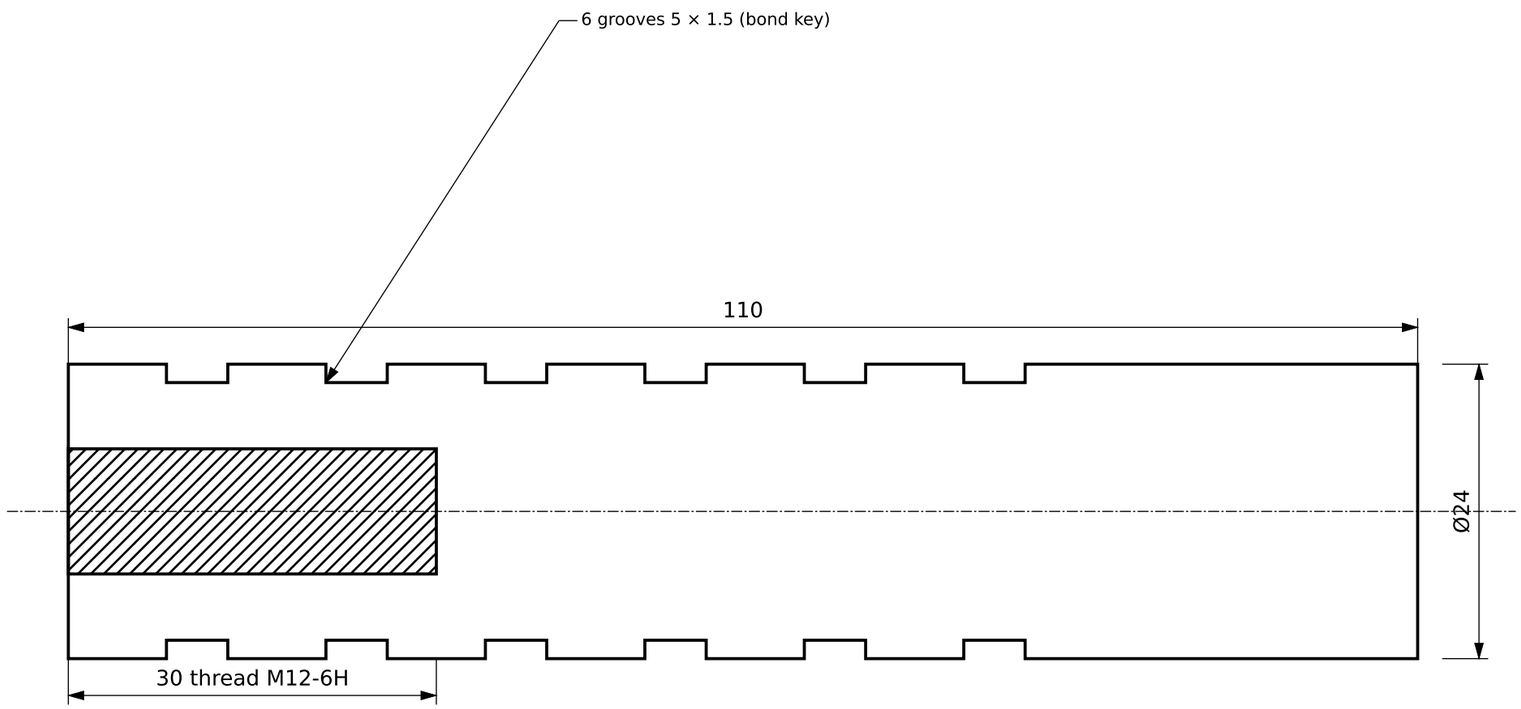


Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

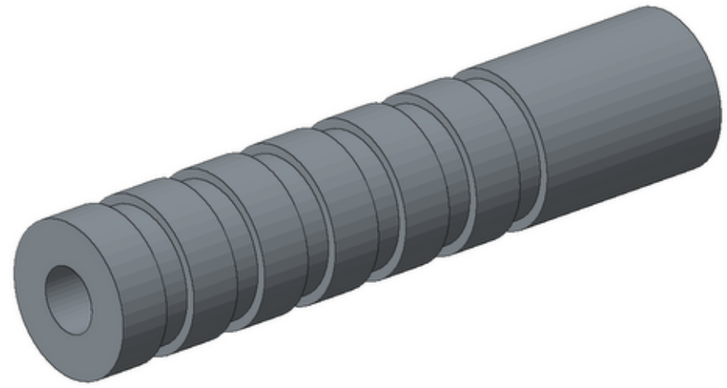
6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

BLADE · SECTIONS AND LAMINATE SCHEDULE

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-101	B · 2026-11-06	1:2 / 1:4
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
E-glass/epoxy (infusion), PVC H80 core	20.6 kg	Claude / not checked



AXONOMETRIC VIEW (not to scale)



NOTES

1. Bonded into the blade root laminate during lay-up with the root fixture NS6-115: insert axis parallel to the pitch axis within 0.3°, PCD 185 ± 0.2, face flush ± 0.2 after root facing.
2. Pull-out test: 3 inserts per batch in a root coupon, ≥ 60 kN each (design 43 kN at SLM H, doc 30 §2).
3. Threads protected with plastic plugs until assembly; studs M12 10.9 × 75 (NS6-900) screwed in 30 mm with Loctite 270, 40 Nm.

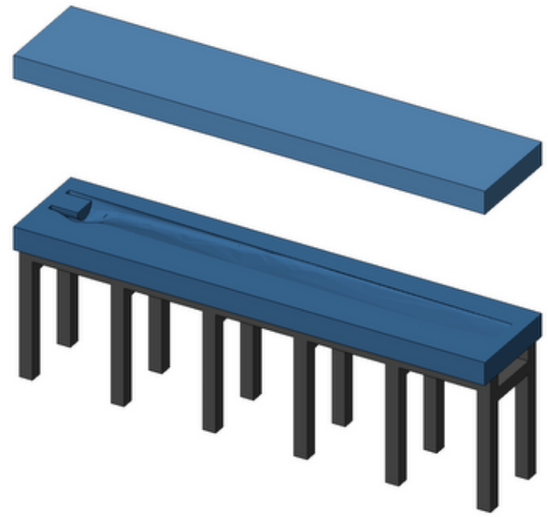
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

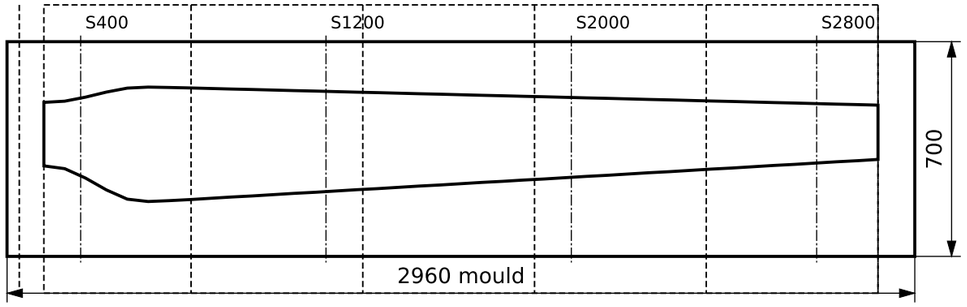
BLADE ROOT INSERT M12 (BONDED)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-105	B · 2026-11-06	2:1
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
C45E+QT, zinc-flake coated (Geomet 500)	0.33 kg	Claude / not checked

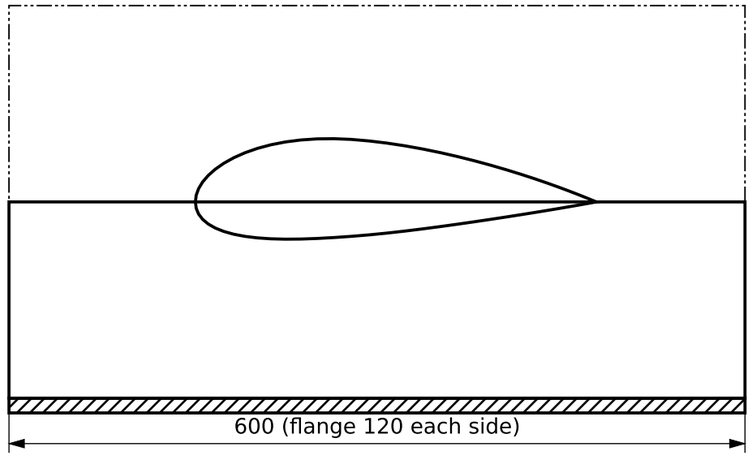
AXONOMETRIC VIEW (not to scale)



PLAN, LOWER HALF (pressure side), frame RHS 80 × 80 × 4 hidden



SECTION S1200: lower half solid, upper half phantom; parting line on LE/TE



Step	Process
1 Plug	CNC-machined tooling board (polyurethane 0.7 g/cm³) on an MDF/steel core; ruled parting surface along LE and TE; sand P600, seal, release
2 Halves	Tooling gelcoat 1 mm, skin 2 × 300 g CSM, 10 mm vinylester/glass build-up, egg-crate ribs 12 mm GRP every 400 mm
3 Frame	RHS 80 × 80 × 4 cradle, 6 legs, levelling feet, pins Ø20 at 4 places on the flanges for closing
4 Heating	silicone heating mats 300 W/m² under the lower half, 70 °C post-cure; thermocouples ×4
5 Web	flat-plate web mould: tapered aluminium plate 2 900 × 220 on the bench (NS6-115)

NOTES

1. One mould set makes all blades (≥ 300 parts before refurbishment). Mould flange 120 mm with a 25 mm vacuum sealing land and a 6 mm resin trap groove.
2. Tolerances: profile ± 0.5 mm, twist ± 0.2°, length ± 2 mm. Check each new mould against the plug with the five section templates S400 … S2800.
3. Root end: 20 mm steel end plate with the bolt pattern of the root fixture NS6-115 (4 × M12), square to the pitch axis.

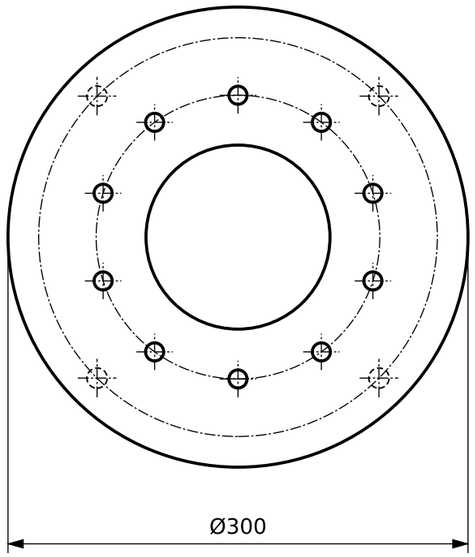
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

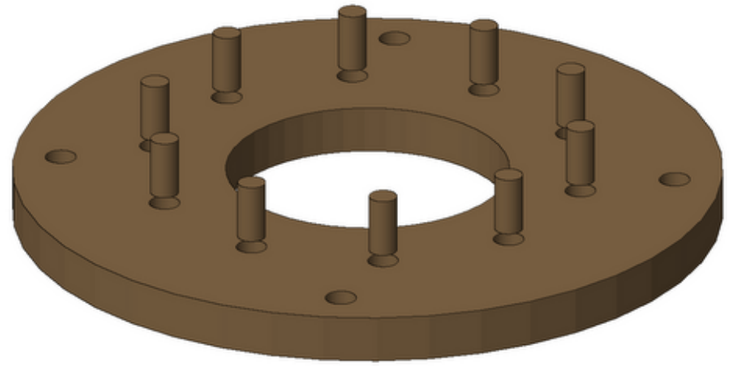
BLADE MOULD SET (PLUG, TWO HALVES, FRAME)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-110	B · 2026-11-06	1:20
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
GRP tooling laminate on S235 RHS frame	≈ 2 × 75 kg + frame 155 kg	Claude / not checked

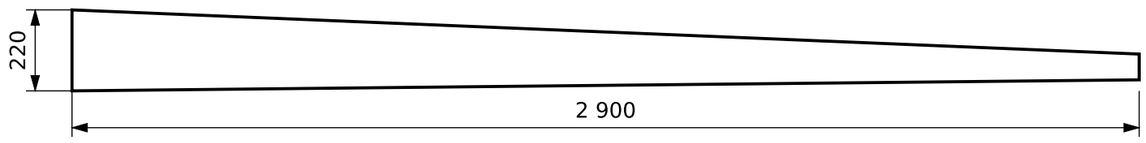
ROOT FIXTURE, view on the spigots: 10 × Ø11.8 g6 spigots × 30 on PCD 185



AXONOMETRIC VIEW (not to scale)



WEB MOULD PLATE (tapered, 10 mm Al)



NOTES

1. Fixture bolts to the root end plate of the lower mould half (4 × M12); inserts NS6-105 slide over the spigots and are screwed on with M12 bolts so their faces sit on the fixture face.
2. After demoulding, remove the fixture, face the root on the root-facing jig (lathe or portable facer) to the root length, check PCD with the hub-flange template.
3. Web mould: flat plate with 25 mm edge flanges; web is infused flat (core 10 mm, 2 × 450 g ±45° each face) and cut to the template lines.

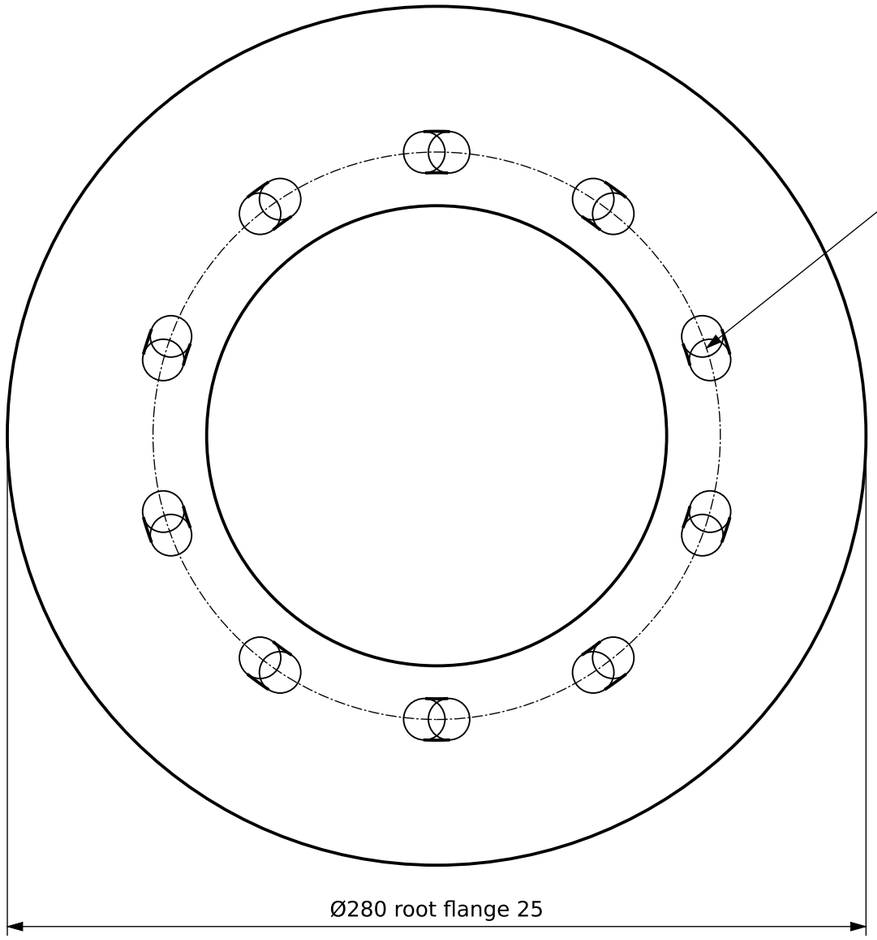
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

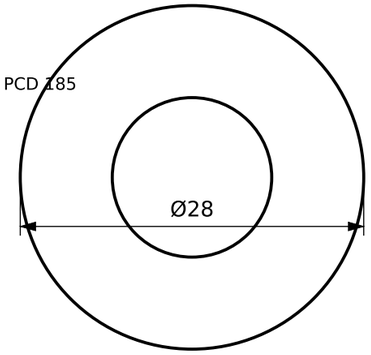
ROOT INSERT FIXTURE AND WEB MOULD (TOOLS)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-115	B · 2026-11-06	1:4
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355 machined; Al 5083 plate	9 kg (fixture)	Claude / not checked

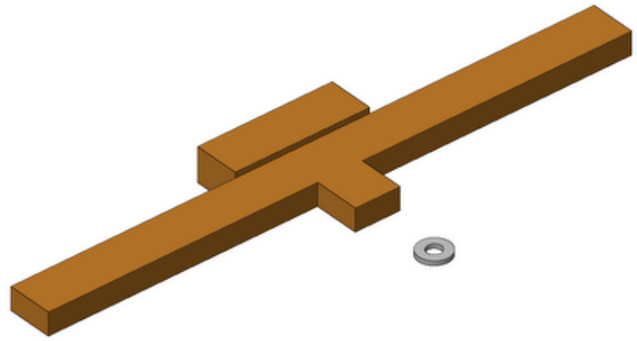
HUB ROOT FLANGE (part of NS6-200), from the blade



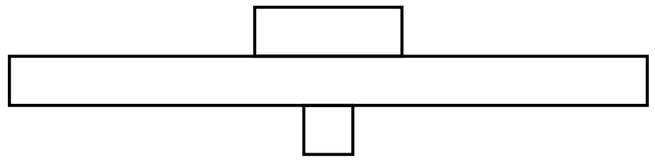
WASHER PLATE 28 × 4, 45 HRC



AXONOMETRIC VIEW (not to scale)



GAUGE NS6-135: clamps on r = 2 000 (chord line pads), inclinometer seat



NOTES

1. Doc 30 called for wedge shims; a wedge tilts the blade axis (cone), it cannot change pitch. Rev B sets pitch by rotating the blade about its axis in arc slots of the hub root flange: ±2.5° around the −2.0° standard (−4.5° … +0.5°).
2. Setting: rotor locked with B1 horizontal, gauge on the blade at r = 2 000, rotor plane reference from a level on the hub front face; set −2.0° ± 0.1°, torque the studs, re-check.
3. Joint: 10 × M12 10.9 studs + hardened washer plates over the slots; friction carries the blade pitching moment (≤ 0.05 kNm) with μ 0.2 ≥ 10× margin.

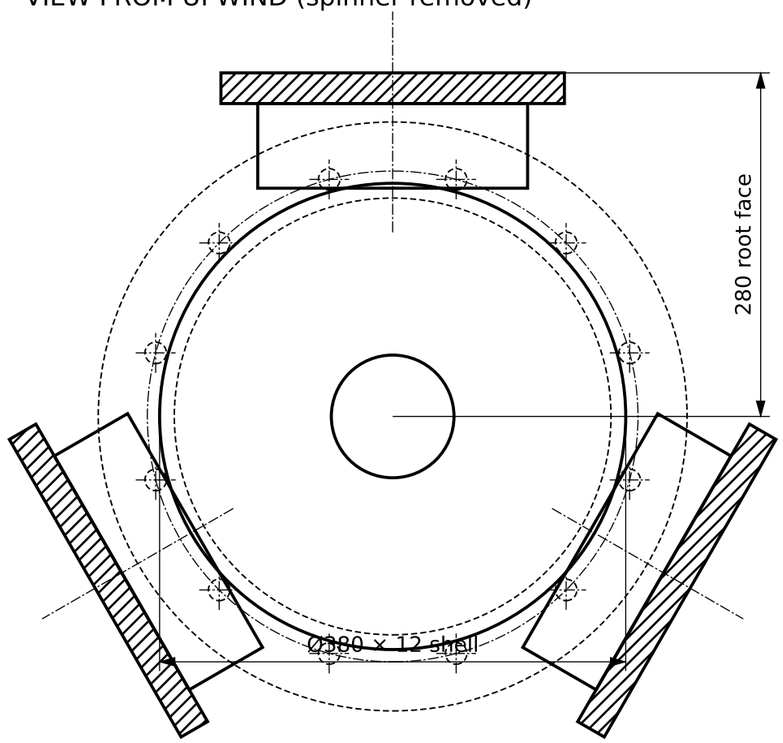
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

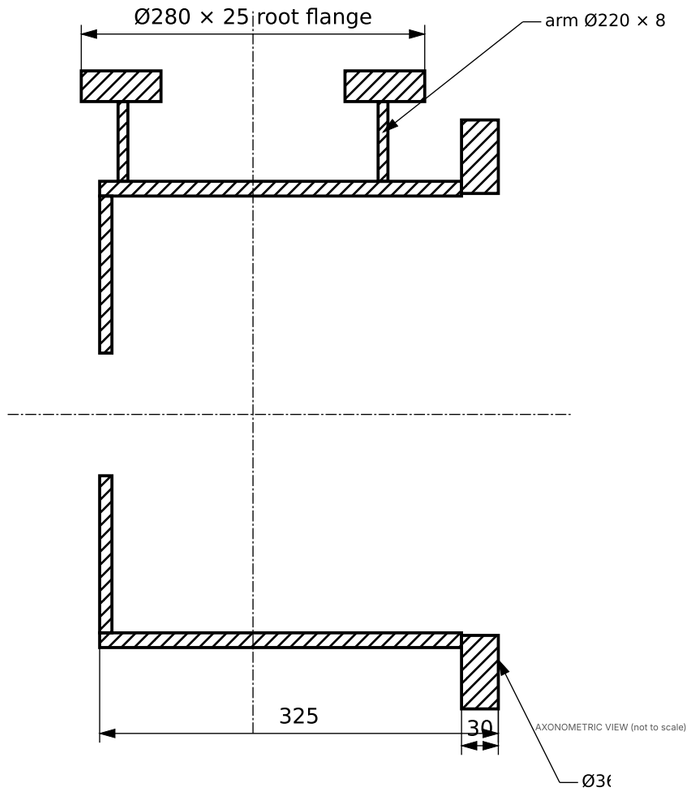
PITCH SETTING · ROOT SLOTS, WASHER PLATES, GAUGE

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-130	B · 2026-11-06	1:2 / 1:5
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
C45 hardened 45 HRC; gauge Al 6082	30 × 15 g; gauge 1.6 kg	Claude / not checked

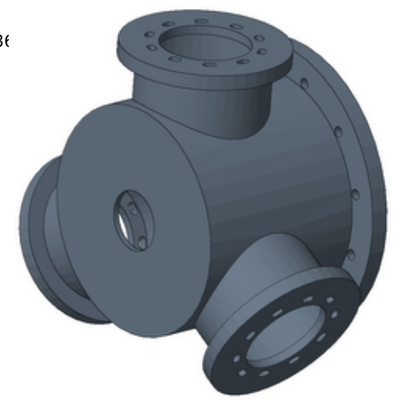
VIEW FROM UPWIND (spinner removed)



SECTION THROUGH THE UPPER ARM (x = 0 rotor plane)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Arms Ø220 × 8: parked SLM stress 68 MPa (allow 323); fatigue at the flange weld 5.7 MPa with simulated loads (allow 30.8, detail 71).
2. Drum joint: Ø360 H7 bore on the drum spigot, 12 × M16 10.9 PCD 400, 250 Nm; friction carries the 5.7 kNm design torque.
3. Root flanges machined after stress relief: face square to the arm axis within 0.05, 10 arc slots ±2.5° on PCD 185 centred on −2.0° pitch (NS6-130); arm 1 marked at the lock-pin azimuth.
4. Shell lengthened to 295 (rev A 240) so the arm intersections close on the shell; front plate 10 with Ø100 access hole. Welds full penetration, ISO 5817 B, 100 % MT, UT on flange welds. Zinc spray + C5 paint.
5. Spinner brackets NS6-211 weld on the front plate edge at 60° between the arms. FE check of the arm-to-shell weld before release (doc 30 §11).

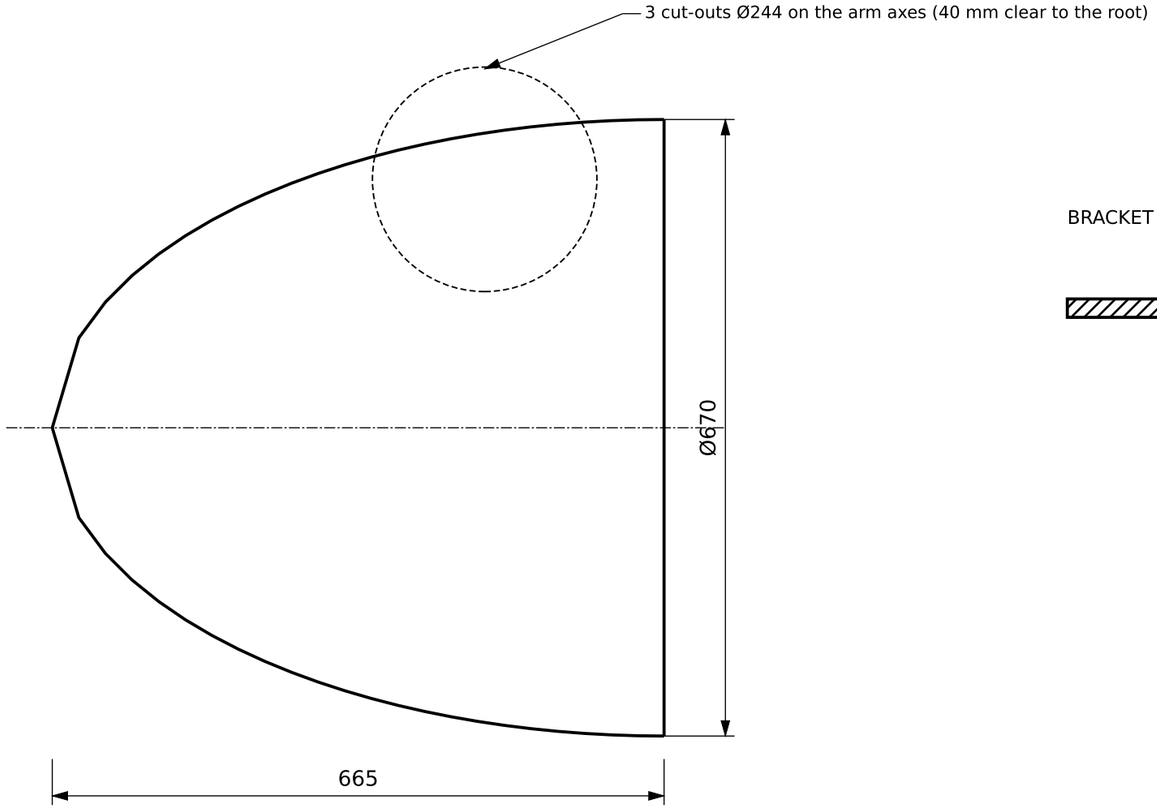
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

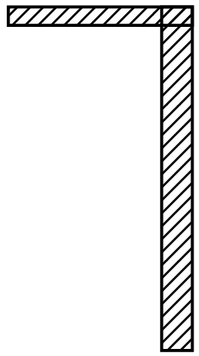
FIXED HUB WELDMENT

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-200	B · 2026-11-06	1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355NL, welded, stress relieved	79 kg (CAD)	Claude / not checked

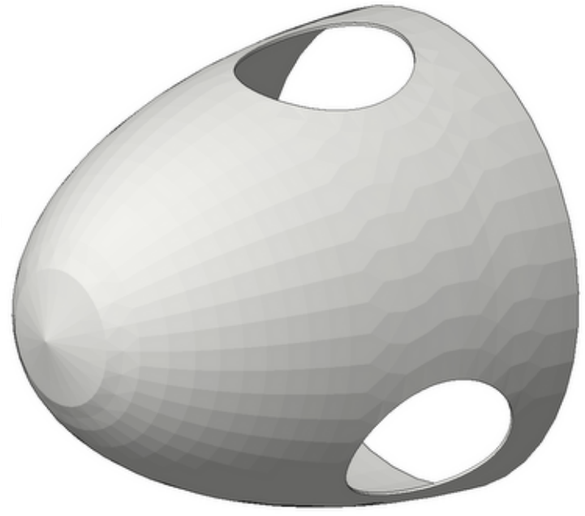
SIDE VIEW



BRACKET NS6-211: 60 × 10 flat, 3 per hub, M8 rivnut in the s



AXONOMETRIC VIEW (not to scale)



NOTES

1. Hand lay-up or RTM in a two-part female mould from the same plug supplier as the blades; 4 mm: gelcoat 0.5 + 3 × 450 g CSM + 1 × 300 g woven; flange-less rear edge with a 20 mm stiffening bead.
2. Rear edge runs 5 mm from the drum front web; the spinner does not seal (drain gaps at the cut-outs).
3. Fix with 3 × M8 1.4404 screws + rubber washers into the brackets; brackets weld to the hub front plate edge at 60° between the arms.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

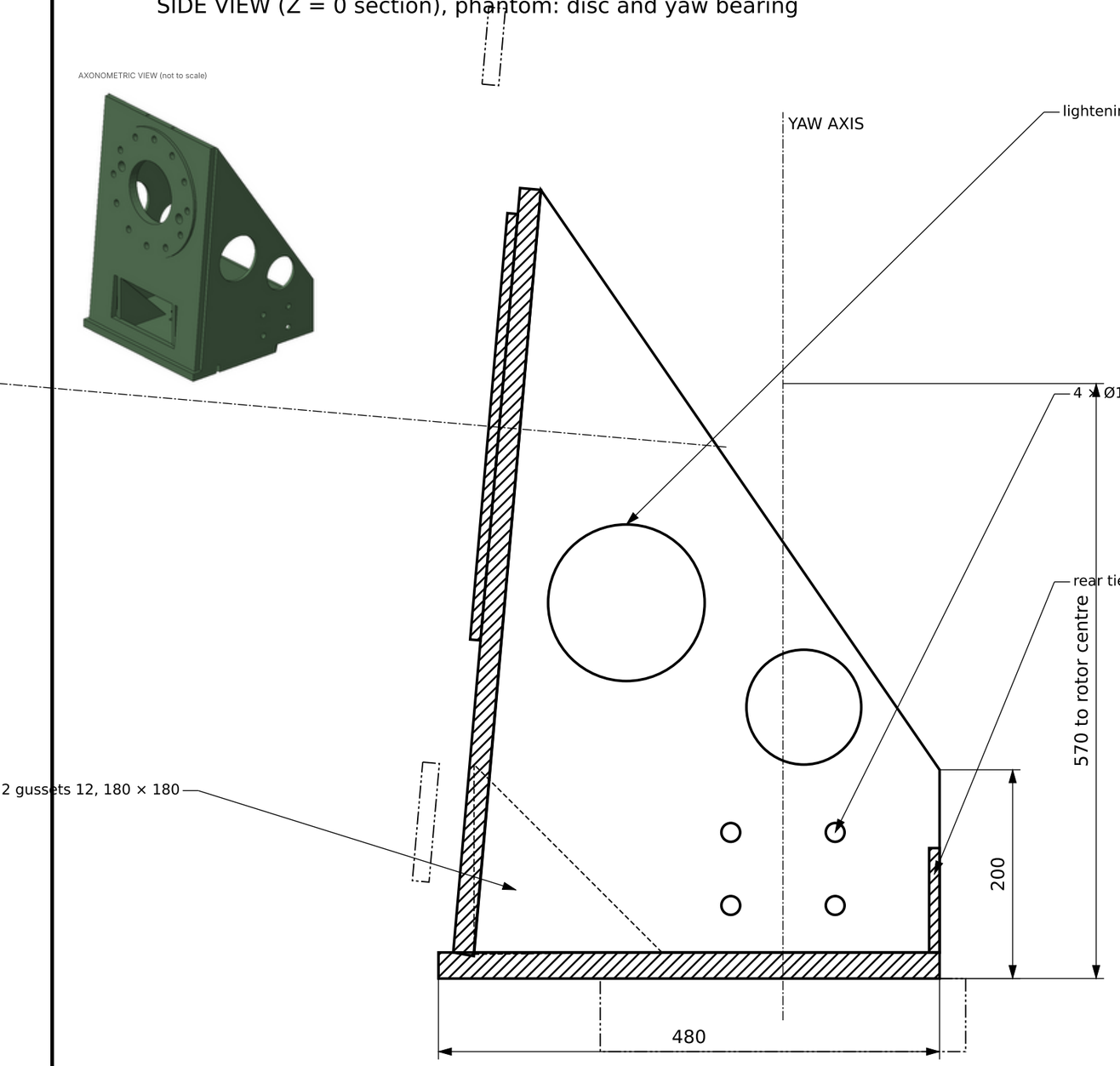
6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

SPINNER (GRP) AND BRACKETS NS6-211

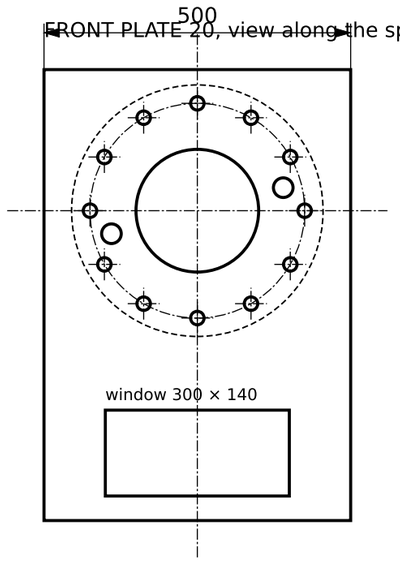
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-210	B · 2026-11-06	1:5
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
glass/polyester 4 mm, gelcoat RAL 9010; brackets S275 galv.	8.7 kg + 3.6 kg	Claude / not checked

SIDE VIEW (Z = 0 section), phantom: disc and yaw bearing

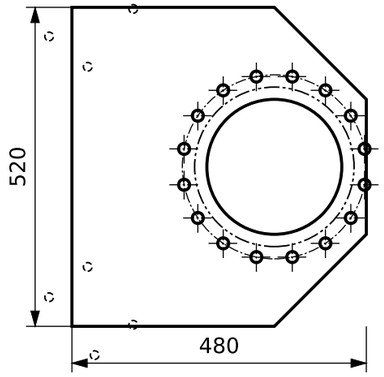
AXONOMETRIC VIEW (not to scale)



FRONT PLATE 20, view along the spindle axis



BOTTOM PLATE (yaw flange), from below



NOTES

1. Replaces NP1-300 (12 kW, 333 kg as drawn) on S6 (doc 32). Same interfaces: Ø410 × 10 pad at x 470, Ø200 H7 spigot, 12 × Ø22 PCD 350 (M20 10.9 from inside, 540 Nm), 2 × Ø32 cable holes PCD 290; bottom Ø220 bore, 16 × Ø18 PCD 300 (yaw inner ring, M16 10.9 with nuts).
2. Plates: front 20, bottom 25, sides 10 (outer faces Y ±260 for NP1-420), rear tie 10, gussets 12. Rear corners of the bottom plate chamfered 150 × 150.
3. Pad units NY-410 bolt under the front corners (4 × M12 each, tapped, hidden); hanger tube flange NY-460 on Ø260 at the bore; caliper bracket NP1-310 on the left side plate front edge; lock pin holder NP1-321 on 2 × M12 in the front plate top edge.
4. Design checks (SLM γ 3, doc 32): root section 21 MPa, front plate 51 MPa, bottom plate 144 MPa (allow 323); fatigue 0.07. Thicknesses are set by machining and slewing ring flatness, not strength: FE stiffness check of the ring seat before release.
5. Welds ISO 5817 B, 100 % MT; stress relieve; machine front and bottom plates in one set-up on a 5° fixture: front face square to the yaw flange within 0.05/400 at 5.00° ± 0.05°; ring seat flatness 0.05.

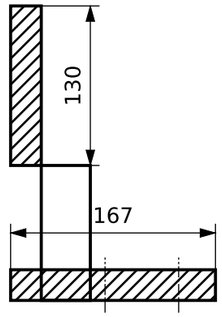
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

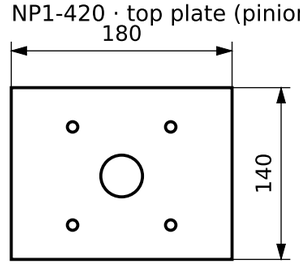
BEDPLATE WELDMENT 6 kW (LIGHT)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-300	B · 2026-11-06	1:5
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N plates, stress relieved, machined	127 kg	Claude / not checked

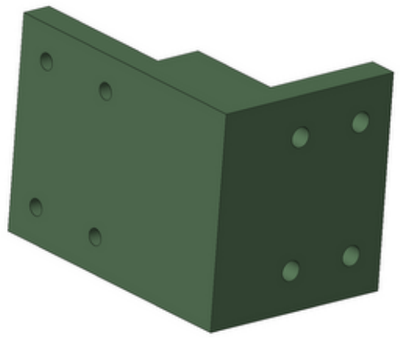
NP1-310 · plan (x along the spindle, y lateral)



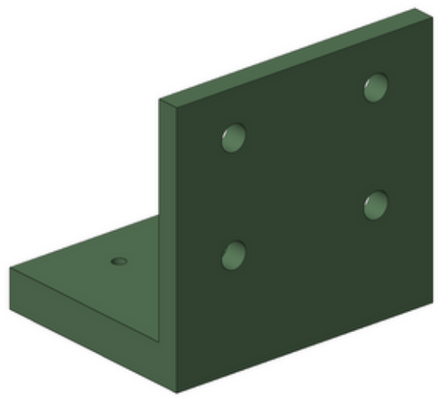
NP1-420 · top plate (pinion bore Ø34, 4 × M10)



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



NOTES

1. NP1-310 (12 kW Rev B geometry, cad/nacelle_fitcheck): only the LEFT bracket is fitted on S6 (one caliper). Mounting plate 25 with 4 × M16 tapped to suit the chosen caliper; web 25 to the bedplate side plate, 4 × M16 10.9 through, 210 Nm.
2. NP1-420 (doc 25): top plate 25 + web 25; web bolts to the side plate outer face with 4 × M16 at x ±50, z 70/140; gearbox flange 4 × M10 8.8, 49 Nm. Centre distance pinion–ring 328.0 ± 0.1 (shim the web).
3. Both: welds a = 8 fillet all round, ISO 5817 C; hot-dip galvanise after welding, then machine the caliper face.

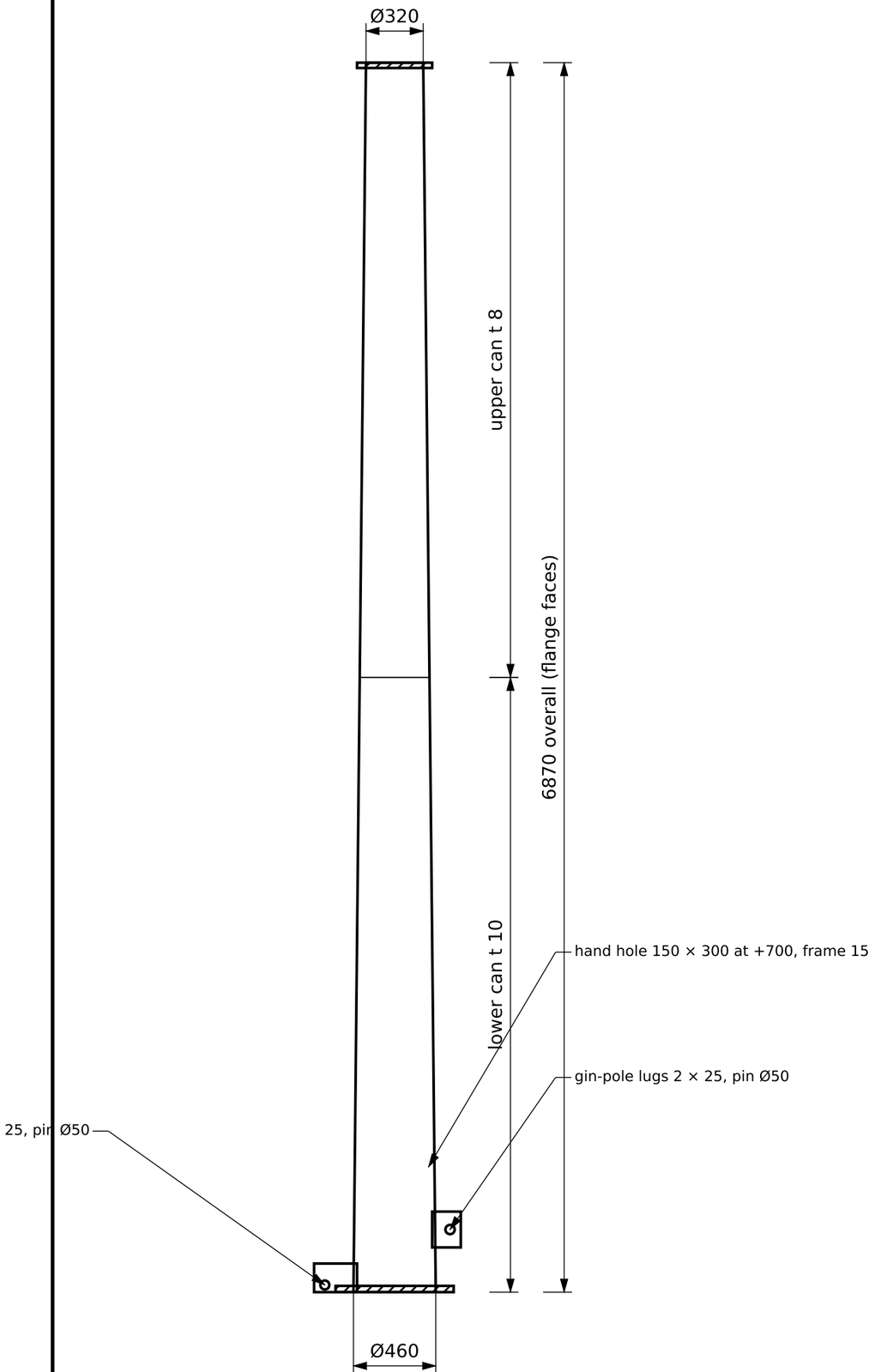
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

CALIPER BRACKET NP1-310 AND YAW DRIVE BRACKET NP1-420

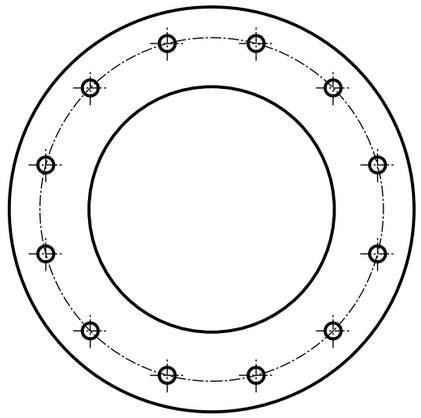
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-310	B · 2026-11-06	1:5
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N plate 25, welded, galv.	18.1 kg; 2 × 7.7 kg	Claude / not checked

ELEVATION

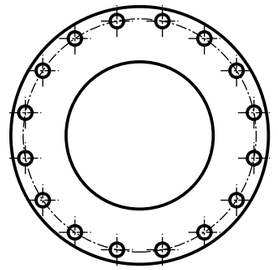


Item	Value
Shell	Ø460 × 10 → Ø320 × 8, linear taper, D/t ≤ 48.3
Cans	2 × 3.44 m rolled, one circumferential butt weld
f1 (rigid / soft+ice)	3.5 / 3.28 Hz; f2 24.04 Hz
Window	f1 ≥ 1.15 × 1P max (3.14 Hz); 3P crosses at 60–77 rpm
Governing	earthquake 0.5 g (ground D) + operation, use 0.445
Base moment (design)	238.6 kNm; SLM sensitivity use 0.601
Base bolts	12 × M24 10.9 anchors: 184.6 / 254.2 kN
Fatigue (base / mid)	0.047 / 0.0245 (turbulence × 1.234)
Tilt-up	base moment 114.7 kNm, use 0.214

FLANGES (plan)

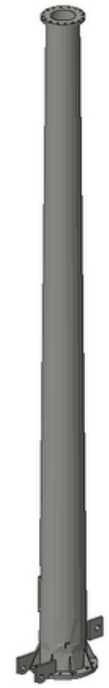


BASE FLANGE 35: 12 × Ø26 PCD 560



TOP FLANGE 30: 16 × Ø22 PCD 380 (to NY-400)

AXONOMETRIC VIEW (not to scale)



NOTES

1. Rev B replaces the 18 m two-section tower. At 8 m a soft tower (f1 below 3P at 41 rpm) is impossible with the 983 kg head, so the tower is stiff: f1 above 1P at full speed; the controller skips the 3P crossing zone like the old 1P zone.
2. Seam and can welds full penetration, ISO 5817 B, 100 % UT on flange and can welds; flanges machined flat after welding (0.2 mm), top flange square to the axis within 0.5 mm/m.
3. No climbing: service by tilting the tower down (NS6-420). Earthing bosses 2 × M12 at +300. Hand-hole cover NS6-401 with EPDM gasket and 6 × M8.

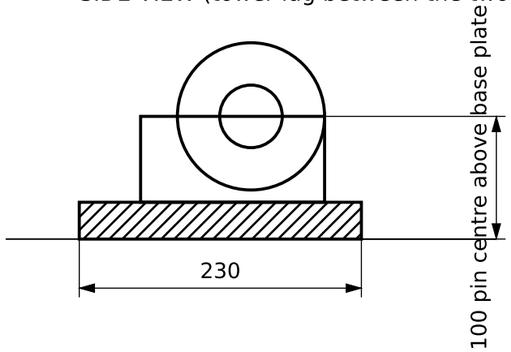
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

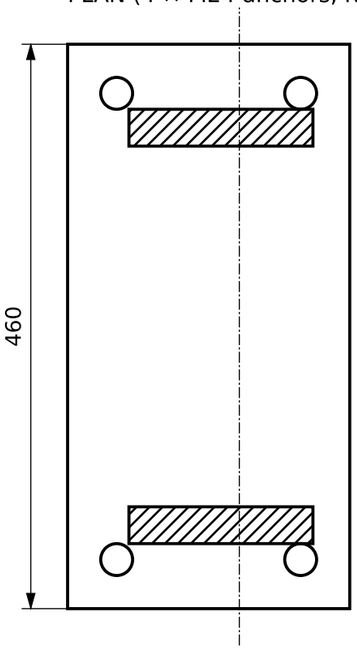
TOWER 8 m · ONE-PIECE TILT-UP MONOPOLE

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-400	B · 2026-11-06	1:25 / 1:10
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N, rolled and seam welded; hot-dip galvanized	681 kg (CAD 698)	Claude / not checked

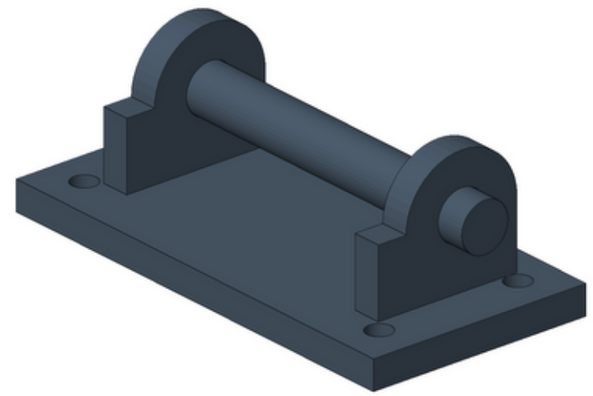
SIDE VIEW (tower lug between the two bracket plates)



PLAN (4 × M24 anchors, NS6-510)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Hinge axis 390 mm upwind of the tower axis (−X side), 40 mm above the base-flange underside: the tower lands on the levelling nuts as it reaches vertical.
2. Pin Ø50 h9 in Ø51 holes, secured with R-clips; pin load at tilt-up 2 × 55 kN shear (double shear, use 0.15).
3. Base plate 30 sits on the pedestal under the grout line; anchors 4 × M24 8.8, 350 Nm after the grout has cured.

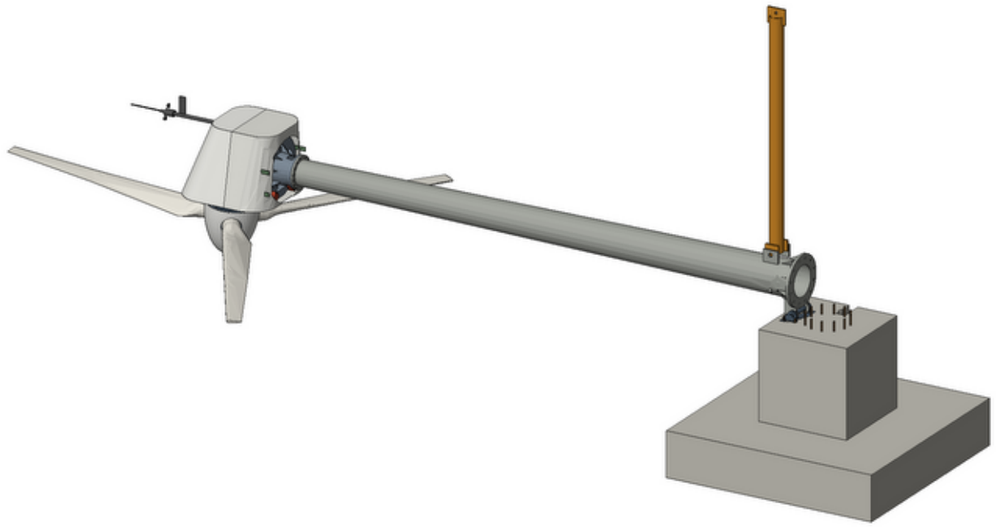
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

BASE HINGE BRACKETS AND PIN

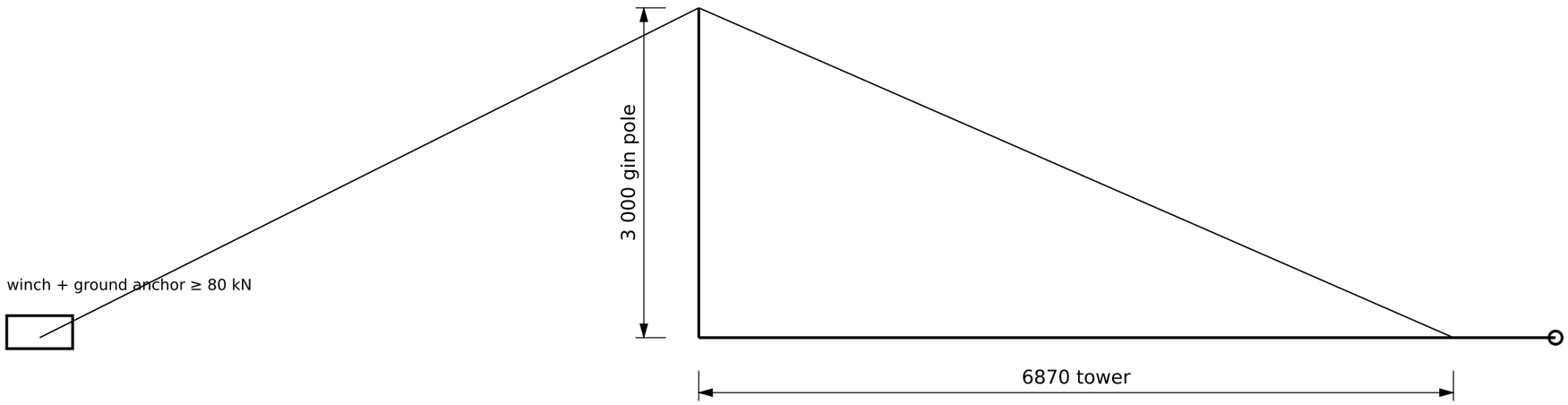
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-410	B · 2026-11-06	1:5
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N 30 mm plate, galv.; pin 42CrMo4+QT Ø50 h9	37 kg	Claude / not checked

AXONOMETRIC VIEW (not to scale)



Item	Value
Gin pole	3.0 m, fixed square to the tower on its lugs
Peak line pull	≈ 38 kN (wind ≤ 10 m/s)
Bridle	2 × wire rope Ø16 6×36 IWRC to the top flange lugs
Pull line	Ø16 wire rope, winch ≥ 50 kN with 1:2 block
Ground anchor	screw anchors or concrete block, ≥ 80 kN at 6 m
Alternative	truck crane ≥ 3 t at 9 m reach (lift by the top flange)

TILT-UP (tower horizontal on the hinge, head fitted, rotor locked)



NOTES

1. Two people + winch operator; wind below 10 m/s; nobody inside the fall zone (1.5 × tower + rotor). Rotor locked with pin NP1-320, blade B1 pointing up away from the ground.
2. Lower the tower the same way for service; support the head on a trestle 1.0 m high so the lower blade clears the ground.
3. After erection: fit base nuts and torque (NS6-900), remove the gin pole, keep it with the site kit.

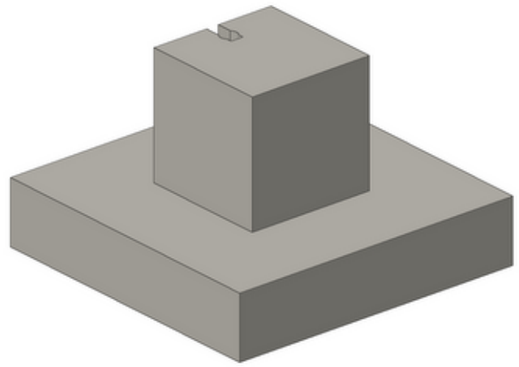
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

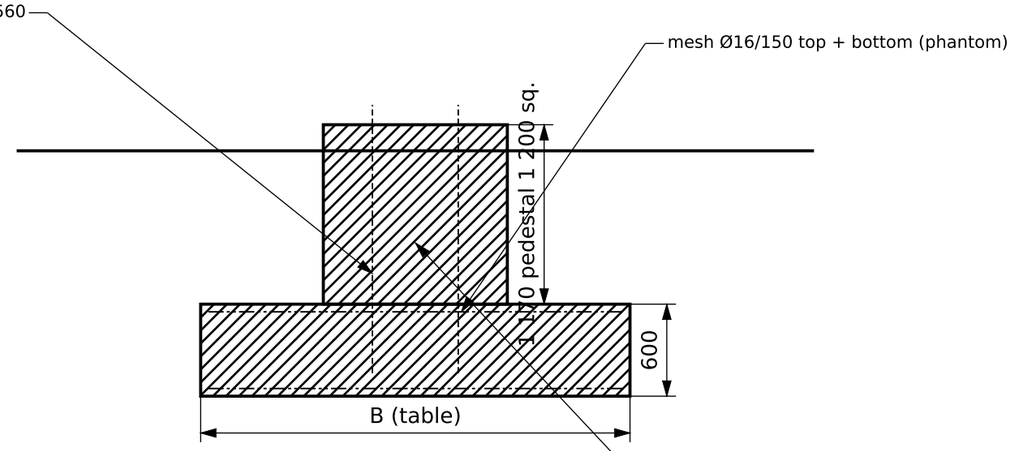
GIN POLE AND TILT-UP RIGGING (ERECTION TOOL)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-420	B · 2026-11-06	1:50
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355 tube 168.3 × 6.3, plates 25	94 kg	Claude / not checked

AXONOMETRIC VIEW (not to scale)

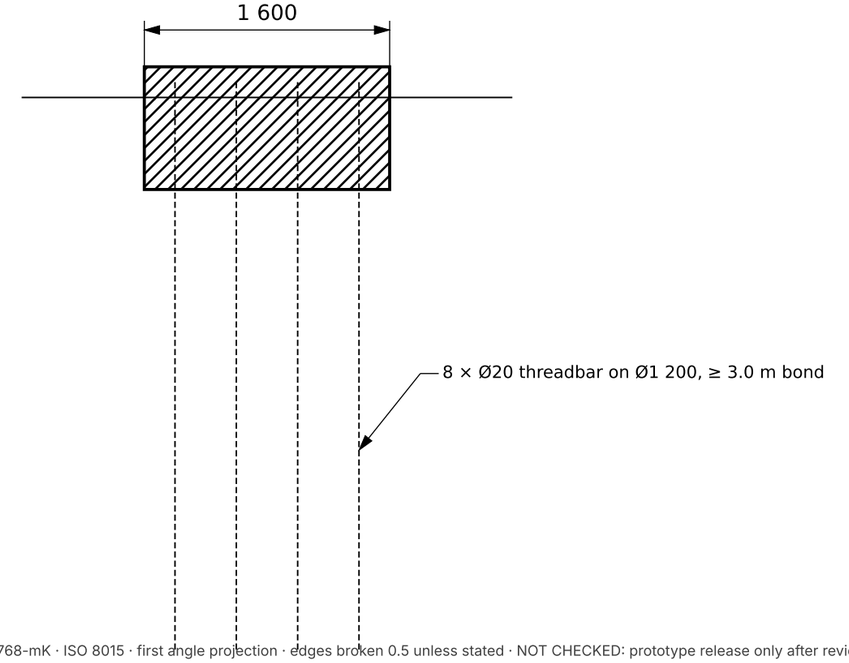


SPREAD FOOTING, SECTION



Ground	B m	Concrete m³	Bearing kPa	q_rd kPa
firm soil (dense gravel, moraine)	2.8	5.5	178.6	300.0
soft volcanic soil (loose ash, pumice sand)	3.1	6.6	99.0	100.0
frost-susceptible silt (thawed)	2.9	5.9	135.5	150.0
rock (anchored plinth)	1.6	2.0	–	anchors 92.1 kN

ROCK PLINTH (lava, basalt)



NOTES

1. Base 1.6 m below ground (frost depth); pedestal 1.2 × 1.2 m to +0.17, grout 30 to +0.20 under the base flange. Design base moment 238.6 kNm (earthquake 0.5 g + operation governs), parked V_e50 175.3 kNm.
2. Characteristic eccentricity ≤ B/4, operation ≤ B/6, bearing on the effective area ≤ q_rd (EN 1997 approach 2 style), as doc 09.
3. Concrete C30/37 XF4 air-entrained, cover 50; mesh Ø16/150 both ways top and bottom, pedestal 12 × Ø16 verticals + Ø10/150 links; earthing ring 50 mm² Cu round the footing, 2 rods 3 m.
4. Soil class to be confirmed by a site inspection; soft volcanic ash needs a plate-load test. Rock: drill Ø76, grout, proof-load each anchor to 1.25 × design.

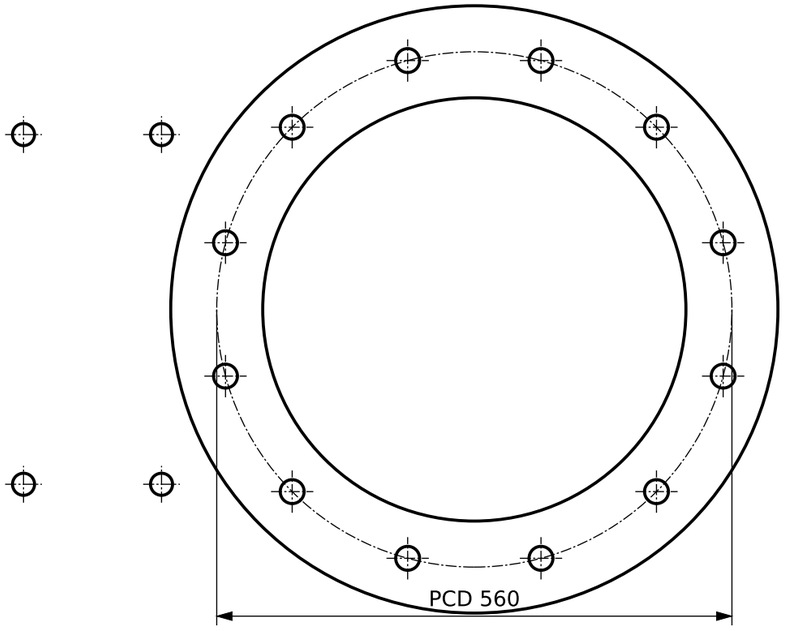
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

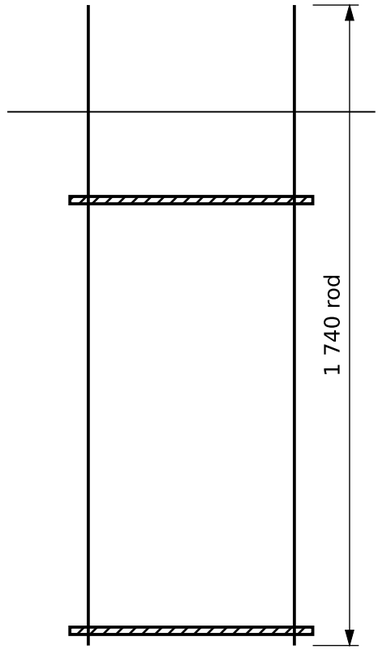
FOUNDATION · SPREAD FOOTING AND ROCK PLINTH (8 m)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-500	B · 2026-11-06	1:40
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
C30/37 XF4 air-entrained; B500C rebar	5.5–6.6 m³ (soil), 2.0 m³ (rock)	Claude / not checked

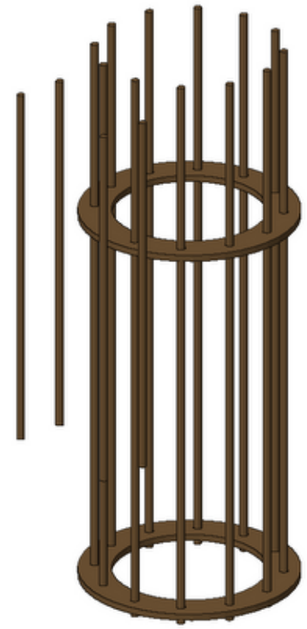
PLAN: template ring (top) on the anchor cage; 4 hinge anchors at −X



ELEVATION (ground at 0)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Anchor rods M24 8.8 HDG × 1 740, rolled thread both ends, 2 nuts + plate washer at the bottom ring; top 260 above the pedestal: levelling nut, base flange 35, washer, nut, lock nut.
2. Set the cage with the template ring (= base flange drilling) bolted on top; position ± 3 mm, level ± 1 mm; the hinge anchors from a second template bolted to the same ring.
3. Pretension after the grout (≥ 40 MPa) has cured 7 days: 450 Nm (≈ 110 kN) with a calibrated wrench, re-check after 1 month.

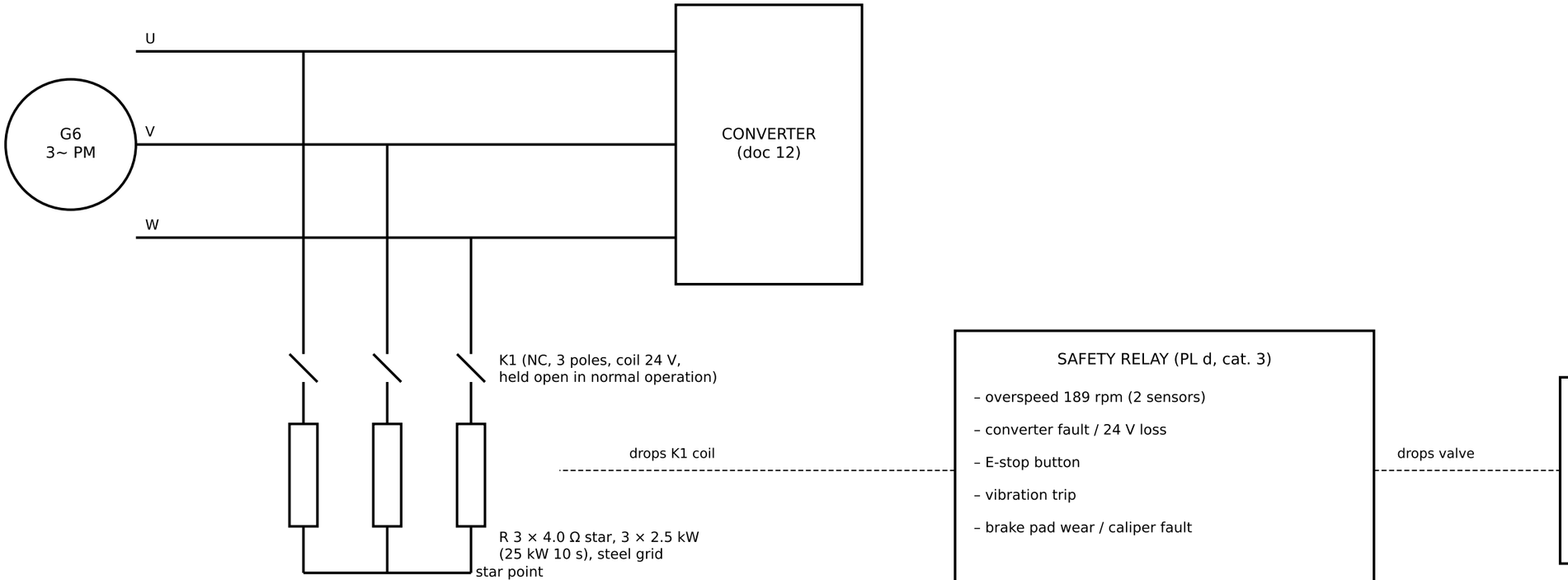
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

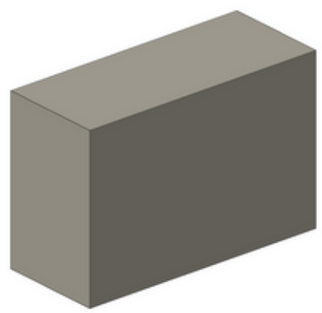
ANCHOR CAGE AND TEMPLATE

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-510	B · 2026-11-06	1:10
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
M24 8.8 HDG threaded rod; S355 rings 100 × 20	144 kg	Claude / not checked

SINGLE-LINE: BOTH STOPPING CHANNELS DROP TOGETHER ON A TRIP



AXONOMETRIC VIEW (not to scale)



NOTES

1. Doc 29 / 31: K1 is normally closed. While the safety relay is healthy it holds K1 open; any trip or loss of 24 V closes K1 and drops the brake at the same time. Either channel alone keeps the rotor below the trip speed (doc 31: brake alone stops in ≤ 2.4 s, peak 173 rpm; resistor alone slows to a 3–15 rpm crawl).
2. Resistor current 43 A peak at trip speed with hot magnets; energy of one stop ≈ 120 kJ. Grid resistors in a ventilated 1.4301 box (600 × 400 × 250) on the tower base cabinet wall, not in the nacelle.
3. K1: 3-pole contactor with mechanically linked mirror contact (EN 60947-4-1 Annex F) for monitoring; AC-3 rating ≥ 63 A at 690 V.
4. Brake fault alarm: with the brake failed the rotor only crawls, so a failed or worn brake keeps the turbine parked and yawed out until repaired (doc 31 §6).
5. Spring pack stays 1.5 kNm (Martin, 2026-11-06). Parked yaw-lost 50-year peak aero torque 1.45 kNm; if exceeded the rotor creeps slowly against the brake and the closed K1 (doc 32 §1). Use the lock pin NP1-320 for long unattended outages.

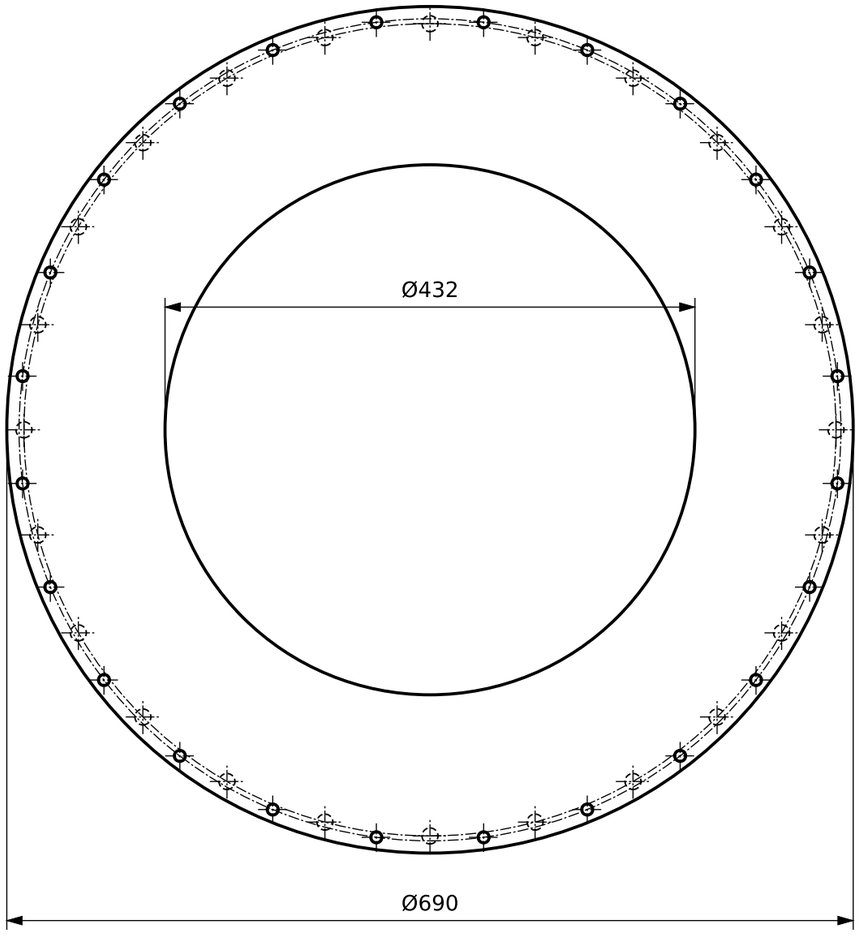
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

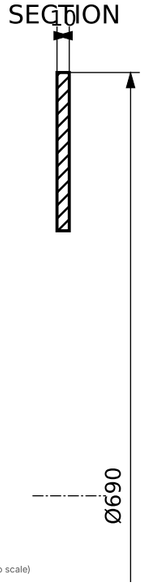
BRAKING RESISTOR BOX AND SAFETY CHAIN

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-600	B · 2026-11-06	schematic / 1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
Box 1.4301, IP54; resistors steel-grid	≈ 18 kg	Claude / not checked

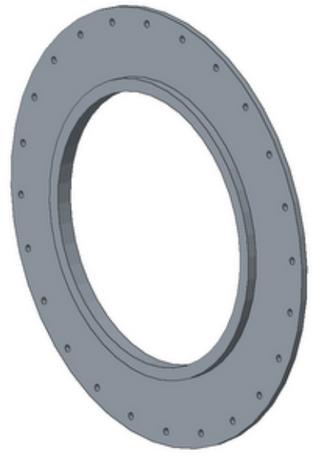
VIEW FROM THE REAR



SECTION



AXONOMETRIC VIEW (not to scale)



NOTES

1. Replaces the Rev C rear cover with 24 fan blades (doc 23/29): G6 runs at half load with no forced cooling; hot spot 86 °C at 1.2 × rated torque without fins, ribs or blades (doc 29).
2. 24 × M8 on PCD 670 to the drum rear ring; seal carrier as Rev C (NP1-212 inner ring).
3. Rear face runs 5 mm from the bedplate front plate outside the Ø410 pad (as Rev B).

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

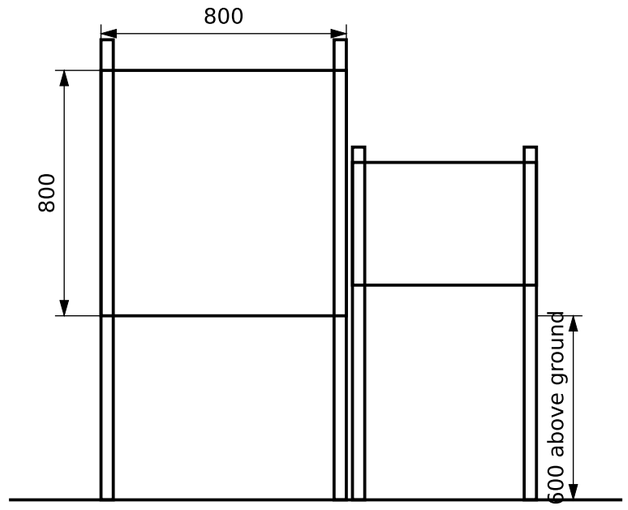
6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

G6 PLAIN REAR COVER (NO FAN BLADES)

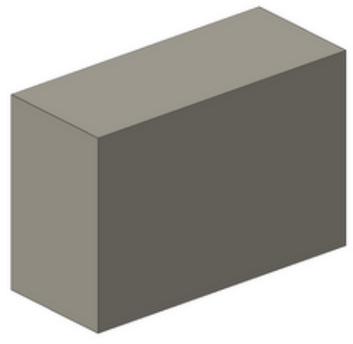
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-610	B · 2026-11-06	1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
EN AW-5083-H111, 10 mm plate	≈ 7 kg	Claude / not checked

Cabinet content (doc 12 modules)	Note
SiC active rectifier 1 200 V, 750 V DC link	6 kW firmware limits
Dump chopper 12 kW + heater feed	hardware trigger 820 V
AC module 6.5 kW (grid / heater)	single- or three-phase
48 V battery module + 200 Wh aux battery	yaw-out without grid
Safety relay PL d, K1 contactor	NS6-600 chain
Controller, modem, surge arresters T1+T2	earth bar to the tower lug

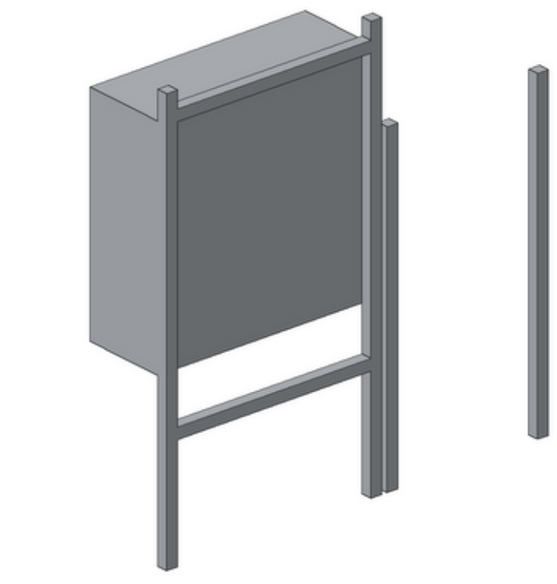
FRONT (from +Y): cabinet left, braking resistor box right (vented)



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Stand on two concrete pads 400 × 400 × 600 (or bolted to the pedestal side); cabinet door faces away from the tower; minimum 1.0 m from the tower axis, outside the rotor disc projection is not required (tip at 5 m).
2. Resistor box NS6-600 mounted 20 mm proud of the stand, louvres down, ≥ 300 mm from the cabinet (heat).
3. Cables enter from below through the two PE ducts in the footing (NS6-800).

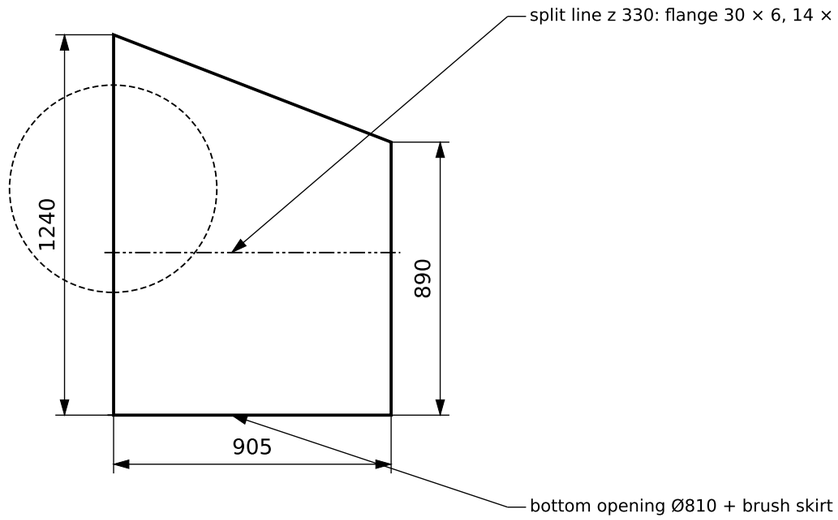
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

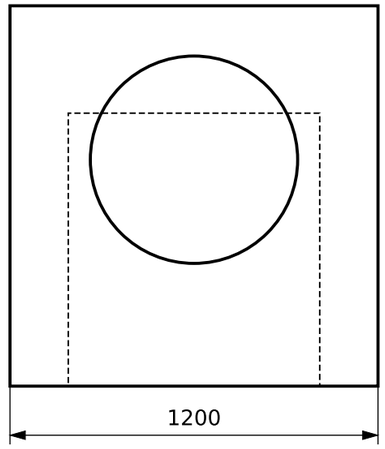
TOWER BASE CABINET, STAND AND RESISTOR BOX POSITION

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-620	B · 2026-11-06	1:20
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
enclosure 1.4301 IP55; stand S235 galv.	95 kg + 18 kg	Claude / not checked

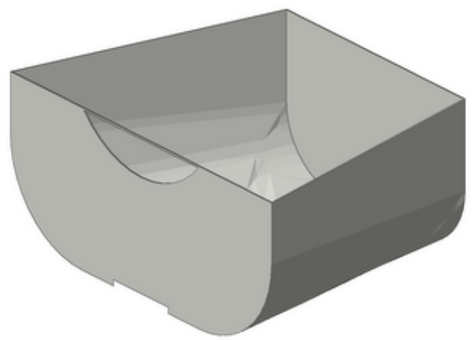
SIDE VIEW (yaw axis at x 0)



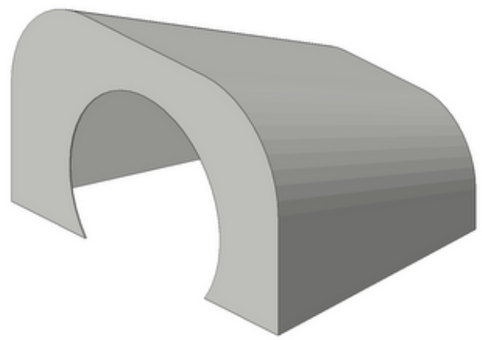
FRONT (opening Ø676 round the drum, 15 gap)



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Lofted rounded-rectangle shell, front 1 200 × 1 240 (R330) to rear 820 × 890 (R230); encloses the disc, caliper, bedplate, yaw drives and pad units; drum and spinner stay outside (drum cools in the free air).
2. Lay-up 4 mm: gelcoat 0.5 + 3 × 450 g CSM + 1 × 300 g woven; PU foam hat-section ribs every 300 mm; lid has 2 lifting points and a gas-strut-held service hatch 500 × 400 over the junction box.
3. Lower tray hangs on the NS6-710 arms (4 × M10 rubber mounts); the lid bolts to the tray flange. Air in at the bottom skirt, out through a labyrinth vent at the rear top (no forced cooling).

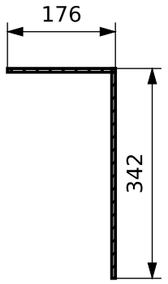
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

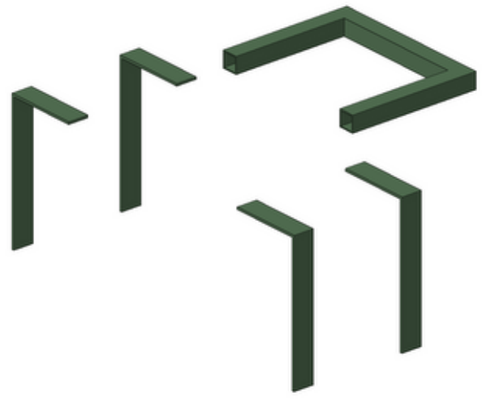
NACELLE COVER · LOWER TRAY AND UPPER LID (GRP)

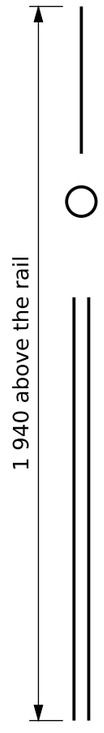
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-700	B · 2026-11-06	1:20
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
glass/polyester 4 mm, gelcoat RAL 9010, PU foam ribs	≈ 22 kg	Claude / not checked

ARM (4 off): flat 50 × 6, bolted 2 × M10 to the side plate



AXONOMETRIC VIEW (not to scale)

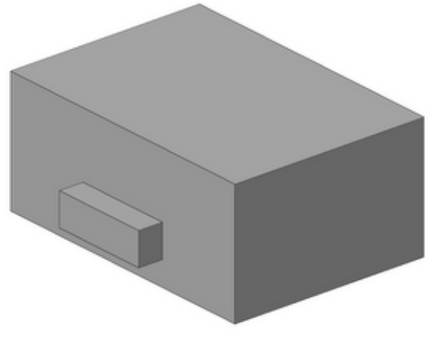




AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Rear rail RHS 40 × 40 × 3 on the bedplate rear tie carries the junction box NS6-730 (IP66, 400 × 340 × 150) and the met mast clamp.
2. Met mast: wind sensors 0.5 m above the cover top (doc 25), air terminal bonded with 50 mm² Cu to the bedplate; the bedplate–tower path goes through the yaw pad brush (NY-410).
3. Junction box: phase terminals 3 × 16 mm², brake/sensor terminals, surge arresters for the sensor lines, 24 V supply; cable glands at the bottom only.

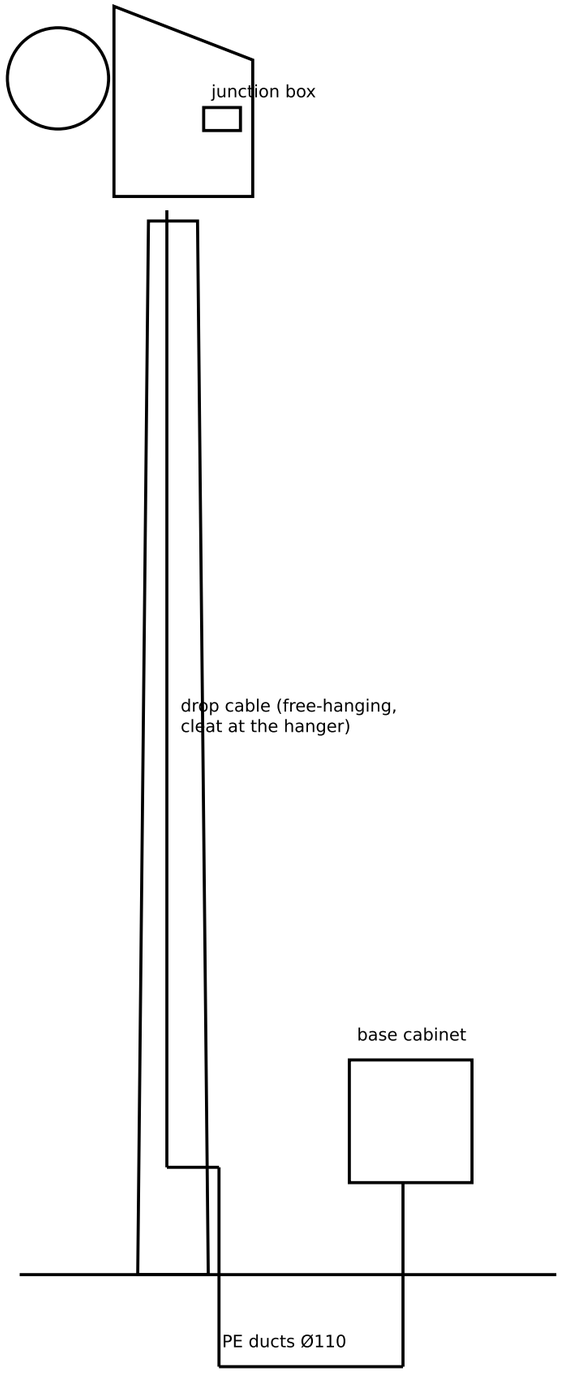
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

COVER ARMS, REAR RAIL, MET MAST NS6-720, JUNCTION BOX NS6-730

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-710	B · 2026-11-06	1:10
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S235 galv.; mast 1.4404 Ø42.4 × 2	22 kg; 4.5 kg; 9 kg	Claude / not checked

ROUTE (nacelle → loop → tower → cabinet)

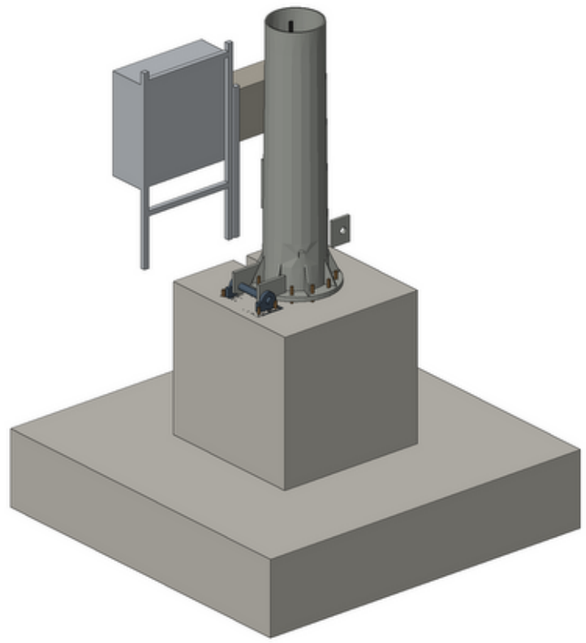


Cable	Type	From → to	Length m
W1 power	H07RN-F 4G10 (3 ph + PE)	generator terminals → junction box	3
W2 power	H07RN-F 4G10, 2.5 turns loop	junction box → hanger loop → hand hole	12
W3 power	NYY-J 4 × 10 in duct	hand hole → cabinet	4
W4 brake	ÖLFLEX 4G1.5 shielded	caliper → junction box → cabinet	16
W5 sensors	LiYCY 12 × 0.5	overspeed, encoder, cam, met, PT100 → cabinet	16
W6 yaw motors	ÖLFLEX 7G1.5	2 gearmotors + brakes → cabinet	16
W7 earth	Cu 50 mm² green/yellow	air terminal → bedplate; tower lug → earth ring	6
W8 data	CAT6 outdoor or fibre	cabinet → house / grid meter	site

NOTES

1. Cable loop NY-460: power and control cables hang free in the tower below the cleat with 2.5 turns of slack; cable-twist limit ±2.5 turns from the encoder/cam switch (doc 25), auto-untwist when idle.
2. Lightning (LPL I, doc 02): air terminal → bedplate → yaw pad brush → adapter → tower → base flange → 2 earthing lugs → ring earth (≤ 10 Ω). Generator winding star point not earthed (IT, insulation monitor in the cabinet).
3. Surge protection: T1+T2 arresters at the cabinet entry for W2/W3, data-line arresters for W5/W8; shields bonded at both ends via the junction box gland plate.
4. All cables UV and cold resistant to −40 °C; drip loops at the hand hole; glands IP68; label every core at both ends.

AXONOMETRIC VIEW (not to scale)



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6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

CABLING ROUTES, EARTHING AND LIGHTNING

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-800	B · 2026-11-06	schematic
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see cable schedule	–	Claude / not checked

Joint	Fastener	Qty	Grade / finish	Tightening	Notes
Blade root → hub flange	stud M12 × 75 + nut + washer plate	30	10.9, zinc-flake	95 Nm + 90°	Loctite 270 in the insert, 40 Nm
Hub → drum web	M16 × 70 ISO 4017	12	10.9, zinc-flake	250 Nm (μ 0.12)	check after 50 h
Spinner → brackets	M8 × 20 + rubber washer	3	A4-70	12 Nm	
Spindle → bedplate front plate	M20 × 90 ISO 4014	12	10.9	540 Nm	from inside, NP1-100
Stator carrier → spindle	M12 × 45	12	10.9	110 Nm	NP1-200
Brake disc → drum	M12 × 40	12	10.9	110 Nm	NP1-230
Rear cover → drum	M8 × 20	24	A4-80	20 Nm	NS6-610
Caliper → NP1-310	M16 (caliper pattern)	4	10.9	250 Nm	per caliper supplier
NP1-310 → side plate	M16 × 70 + nut	4	10.9 HV	210 Nm	
Lock pin holder → front plate	M12 × 50	2	8.8	80 Nm	
Yaw inner ring → bedplate	M16 × 110 + nut	16	10.9 HV	250 Nm	doc 25
Yaw outer ring → NY-400	M16 × 70	16	10.9	250 Nm	doc 25
Pad unit → bedplate	M16 × 50	8	10.9	250 Nm	NY-410
Gearmotor bracket → side plate	M16 × 60 + nut	8	8.8	195 Nm	NP1-420
Gearmotor → bracket	M10 × 30	8	8.8	49 Nm	
NY-400 → tower top flange	M20 × 90 + nut	16	10.9 HV	450 Nm	through bolts
Cover arms → side plates	M10 × 25	8	A4-70	35 Nm	rubber mounts
Cover lid → tray	M8 × 25 + EPDM	14	A4-70	10 Nm	
Base flange → anchors	M24 nuts (levelling + top + lock)	12	8.8 HDG anchors	450 Nm	after grout cured 7 d
Hinge brackets → anchors	M24 nuts	4	8.8 HDG	350 Nm	
Hand-hole cover	M8 × 20	6	A4-70	12 Nm	EPDM gasket
Cabinet stand → pads	M12 anchors	4	A4-70	40 Nm	

NOTES

1. Torques for lightly oiled zinc-flake threads (μ ≈ 0.12) unless stated; HV sets per EN 14399-4 with the manufacturer's k-class.
2. Mark every torqued bolt with a torque stripe; record values in the installation report. Re-check all structural joints (blade, hub, yaw, tower) after 50 h and then yearly.
3. Never re-use M16 / M20 / M24 HV sets that have been fully tightened; replace studs that show thread damage.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

FASTENER AND TORQUE SCHEDULE

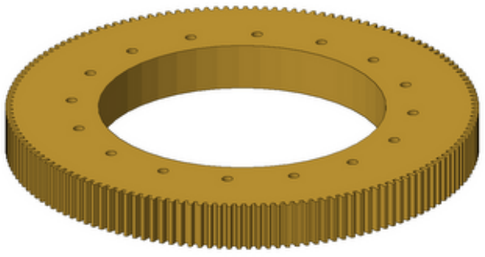
DRAWING NO.	REV / DATE	SCALE (A3)
NS6-900	B · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
ISO 4014/4017, EN 14399-4 (HV), 10.9 zinc-flake unless stated	–	Claude / not checked

Item	Specification	Interface	Candidates / note
NS6-911 yaw ring	4-point ball slewing ring Dpw 360, ext. gear z150 m4, ind. hardened teeth	inner 16 × M16 PCD 300; outer 16 × M16 PCD 440	Rollix / Kaydon / IMO; tilt 34.6 kNm s0 ≥ 1.44
NS6-912 yaw gearmotor (2)	0.18 kW, i ≈ 1 680, output ≥ 1 500 Nm static, brake 2 Nm	flange 4 × M10 on 80 square, pinion z14 m4	SEW / Bonfiglioli / NORD
NS6-913 main bearings (2)	tapered roller 32028 X (140 × 210 × 45)	spindle Ø140 f6 / j6	SKF / FAG / Timken, C3 not needed
NS6-914 overspeed sensors (2)	inductive M18 PNP, −40 °C, IP68	holder NP1-321	ifm / Pepperl+Fuchs
NS6-915 rotor brake caliper	spring-applied, power-released, 16 mm disc, 1.5 kNm at r 0.41 m	4 × M16 on NP1-310	Svendborg BSFI / Altra / Ringspann
Magnets	200 + 10 × N45SH 30 × 15 × 12, NiCuNi + epoxy	drawing NP1-220E	Neo / VAC (non-Asian, doc 23)
K1 contactor	3-pole AC-3 ≥ 63 A 690 V, mirror contacts	NS6-600	ABB / Siemens
Converter, chopper, modules	doc 12 SiC rectifier etc., 6 kW firmware	NS6-620	in-house (doc 12)
Met sensors	heated cup anemometer + vane, −40 °C	NS6-720	Thies / Lufft
Yaw encoder	absolute multi-turn SSI, IP67	NY-440	Kübler / Sick

NOTES

1. Envelopes in the CAD model are estimates; replace them with the supplier STEP files before the clash check of the prototype build.
2. All bought parts: operating −30 °C, survival −40 °C, salt fog C5 (doc 02); certificates 3.1 for structural parts (slewing ring, bearings, bolts).

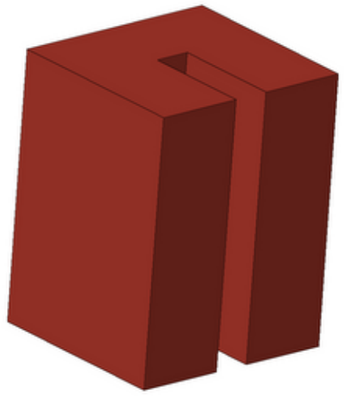
AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



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6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

BOUGHT PARTS SPECIFICATION (ENVELOPES AND INTERFACES)

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-910	B · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see lines	see lines	Claude / not checked